
PHYS 234

Quantum Physics I

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Credit to Dr. Russell Thompson For His Lectures on PHYS 234.

Preface

A majority of the information presented in this note package is transcribed directly from the *PHYS 234: Quantum Physics I* lectures during the Spring 2025 term, taught by Dr. Russell Thompson.

These notes are by no means exhaustive nor are they guaranteed to be error-free, and I do not make any claims about the reliability or accuracy of these set of notes. They are intended primarily for personal use and for those who may benefit from an alternative presentation of the material.

Please use these notes at your own discretion.

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1 Formalism of Quantum Mechanics

1.1 Dirac Notation

■ We begin by developing a concise method of representing the mathematics of quantum mechanics without the dependence of bases. A single particle can be represented by a "bra", $\langle\psi_1|$, and a "ket", $|\psi_2\rangle$, written as

$$\langle\psi_2|\psi_1\rangle \equiv \int \psi_1^*(\mathbf{r})\psi_2(\mathbf{r}) d\mathbf{r}. \quad (1)$$

Here, $\langle\psi_1|$ represents a linear function of vectors, and $|\psi_2\rangle$ represents a vector. In particular,

$$\langle\psi_1| = \int \psi_1^*(\mathbf{r})[\dots] d\mathbf{r}$$

where a bra operates on a ket, $[\dots]$ ($\dots$ denotes any ket), to perform an inner product and return a complex number. In the case above, we can transform a particle from position space to momentum space, $\mathbf{r} \rightarrow \mathbf{p}$,

$$\langle\psi_1|\psi_2\rangle \equiv \int \bar{\psi}_1^*(\mathbf{p})\bar{\psi}_2(\mathbf{p}) d\mathbf{p}.$$

In fact, the transformation $\mathbf{r} \rightarrow \mathbf{p}$ is given by a Fourier transform from $\mathbf{r}$ space to $\mathbf{p}$ space. Dirac notation also follows similar properties to that of typical algebra:

commutative property: $\langle\psi_1|\psi_2\rangle = \langle\psi_2|\psi_1\rangle^*$

distributive property: $\langle\psi_3|\psi_1 + \psi_2\rangle = \langle\psi_3|\psi_1\rangle + \langle\psi_3|\psi_2\rangle$

associative property: $\langle\psi_1|c\psi_2\rangle = c\langle\psi_1|\psi_2\rangle$ and $\langle c\psi_1|\psi_2\rangle = c^*\langle\psi_1|\psi_2\rangle$

for some $c \in \mathbb{C}$. We define the following terms, **orthogonality** and **orthonormality**. Particularly, two wavefunctions, ψ_1 and ψ_2 , are **orthogonal** if

$$\langle\psi_1|\psi_2\rangle = 0.$$

As a normalization condition, an arbitrary wavefunction must follow

$$\int |\psi(\mathbf{r})|^2 d\mathbf{r} = \int \psi^*(\mathbf{r})\psi(\mathbf{r}) d\mathbf{r} = \langle\psi|\psi\rangle = 1. \quad (2)$$

A function with satisfies both of these conditions is considered **orthonormal**.

1.2 Hilbert Space

We know that a vector exists geometrically, independent of its coordinate system. In particular, we can *represent* a vector in terms of a basis set, $\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3\}$, which *span* the space. In Cartesian, 1, 2 and 3 are replaced for x, y and z . This follows for other coordinate systems as well.

The inner product of any two given vectors is always a scalar,

$$\langle\mathbf{v}|\mathbf{u}\rangle = v_1u_1 + v_2u_2 + v_3u_3 + \dots + v_nu_n$$

and the magnitude of a given vector being $\sqrt{\mathbf{v} \cdot \mathbf{v}}$. Let us construct the notion of a **Hilbert space**.

A Hilbert space is the same type of space as a the familiar, geometric space, with the caveat that it is a space for functions rather than discrete element vectors.

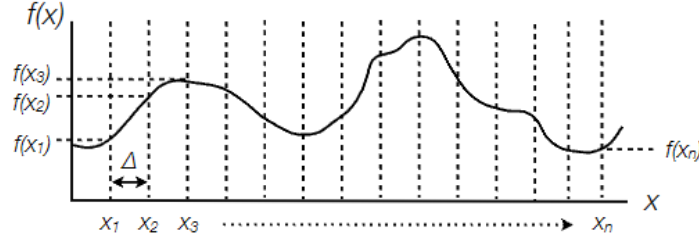


Figure 1: Partitioning a function, $f(x)$, into several values of itself, $f(x_i)$, with spacing $\Delta x = x_{i+1} - x_i$.

We can represent these partitions as $\mathbf{f} = [f(x_1), f(x_2), \dots, f(x_n)]^T$, where the basis of $\mathbf{f}$ is given by $[x_1, x_2, \dots, x_n]$.¹ In the limit that $n \rightarrow \infty$ along with $\Delta \rightarrow 0$, we get the **state function** (wavefunction). Thus, a function is simply a vector in Hilbert space, which in itself is a vector space in infinite dimensions.

The Hilbert space follows particular properties, assuming that ψ exists in $\mathbb{H}$, Hilbert space.

1. $c\psi \in \mathbb{H}$ for $c \in \mathbb{C}$.
2. $\psi_1 + \psi_2 \in \mathbb{H}$.
3. If $\psi_1, \psi_2 \in \mathbb{H}$, then $\langle \psi_1 | \psi_2 \rangle \in \mathbb{H}$.
4. The length of a vector is related to the inner product, $\langle \psi_1 | \psi_2 \rangle$.

1.3 Basis Sets

As is the case in geometric vector space, there are also functions which can span Hilbert space. We denote this as such

$$\{\psi_1(\mathbf{r}), \psi_2(\mathbf{r}), \dots\} \rightarrow \{\psi_n(\mathbf{r})\}$$

and by Dirac notation, we condense this further

$$\boxed{\{|1\rangle, |2\rangle, \dots\} \rightarrow \{|n\rangle\}} \quad (3)$$

Let us discuss specific properties of these spanning functions.

1.3.1 Orthonormality

Recall that orthonormal basis set can be represented as

$$\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij}$$

¹Note this similarity to a geometric vector, $\mathbf{v} = [v_1, v_2, \dots, v_n]^T$ with basis $[\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n]$.

where δ_{ij} is the Kronecker delta, for $i, j = x, y$ or z . In Hilbert space, the equivalent is

$$\int \psi_n^*(\mathbf{r}) \psi_m(\mathbf{r}) d\mathbf{r} = \delta_{nm} = \begin{cases} 1, & \text{if } n = m \\ 0, & \text{if } n \neq m \end{cases}$$

assuming that $\psi_n^*(\mathbf{r})$ and $\psi_m(\mathbf{r})$ create an orthonormal basis set. Recalling (3), we equivalently rewrite this as

$$\langle n|m \rangle = \delta_{nm}.$$

1.3.2 Projections

As for a geometric vector, we can write a function as a linear combination of its components,

$$\psi(\mathbf{r}) = \sum_n c_n \psi_n(\mathbf{r}) \implies \boxed{\psi(\mathbf{r}) = \sum_n c_n |n\rangle}. \quad (4)$$

In particular, the coefficients are **projections** of ψ onto the vectors, $|n\rangle$, such that

$$\boxed{c_n = \langle n|\psi \rangle}. \quad (5)$$

Proof. Expanding (5) in its integral form and substituting (4),

$$c_n = \int \psi_n^*(\mathbf{r}) \psi(\mathbf{r}) d\mathbf{r} = \int \psi_n^*(\mathbf{r}) \left[\sum_m c_m \psi_m(\mathbf{r}) \right] d\mathbf{r} = \sum_m c_m \int \psi_n^*(\mathbf{r}) \psi_m(\mathbf{r}) d\mathbf{r}$$

where the index, m , in the sum from (4) is a dummy index. We assume that $\psi_n(\mathbf{r})$ and $\psi_m(\mathbf{r})$ are orthonormal, as implied by (4), and thus

$$c_n = \sum_m c_m \delta_{nm} = c_{n=m}$$

where $\delta_{nm} = 1$ and the off-diagonals of the δ_{nm} are zero. In Dirac notation, this is equivalently expressed as

$$|\psi\rangle = \sum_n c_n |n\rangle \implies \langle m|\psi\rangle = \langle m| \sum_n c_n |n\rangle$$

where we have acted $\langle m|$ on both sides. Then,

$$\sum_n c_n \langle m|n\rangle = \sum_n c_n \delta_{nm} = c_{n=m}.$$

□

1.3.3 The Closure Relation

An orthonormal set, $\{|n\rangle\}$, is considered a basis set if, $\forall \psi(\mathbf{r}) \in \mathbb{H}$, $\psi(\mathbf{r})$ can be expressed as an expansion in $\{|n\rangle\}$. Specifically, this is given by the **Closure Relation**²,

$$\boxed{\sum_n \psi_n^*(\mathbf{r}) \psi_n(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}')} \quad (6)$$

²Physically, the Closure Relation says that we are able to, at all points of $\psi(\mathbf{r})$, multiply by $\psi_m^*(\mathbf{r})$ to reproduce a spike at the point $\mathbf{x} = \mathbf{r} - \mathbf{r}'$. Since we do not want $\delta(\mathbf{r} - \mathbf{r}')$ to do more than simply acting as an index, it is necessary that $\delta(\mathbf{r} - \mathbf{r}') = 1$.

where $\delta(\mathbf{r}-\mathbf{r}')$ is the Dirac delta function centered at $\mathbf{r}'$. Alternatively, the equivalent representation in Dirac notation of the Closure Relation is by the **Identity Operator**,

$$\boxed{\sum_n |n\rangle \langle n| = \mathcal{I}.} \quad (7)$$

We note that if the set, $\{|n\rangle\}$ is a basis set, that (6) or (7) *must* be satisfied. Let us determine the rationale behind (6) and (7).

Proof. By, (4) and (5), we have that

$$\psi(\mathbf{r}) = \sum_m c_m \psi_m(\mathbf{r}) = \sum_m \left[\int \psi_m^*(\mathbf{r}') \psi(\mathbf{r}') d\mathbf{r}' \right] \psi_m(\mathbf{r}) = \int \psi(\mathbf{r}') d\mathbf{r}' \left[\sum_m \psi_m^*(\mathbf{r}') \psi_m(\mathbf{r}) \right] \quad (1.P1)$$

where

$$c_m = \int \psi_m^*(\mathbf{r}') \psi(\mathbf{r}') d\mathbf{r}'.$$

However, by (6), $\sum_m \psi_m^*(\mathbf{r}') \psi(\mathbf{r})$, and so

$$\psi(\mathbf{r}) = \int \psi(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') d\mathbf{r}'. \quad (1.P2)$$

Notice that (1.P2) and (1.P1) must be equivalent statements, and by inspection, shows that

$$\sum_m \psi_m^*(\mathbf{r}') \psi_m(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}')$$

as before. In Dirac notation, we can condense this to simply a few lines.

$$|\psi\rangle = \sum_n c_n |n\rangle = \sum_n \langle n|\psi\rangle |n\rangle = \sum_n |n\rangle \langle n|\psi\rangle.$$

It follows that to get $|\psi\rangle$ back from any arbitrary operator, $\hat{Q}$, it must be true that

$$\sum_n |n\rangle \langle n| = \mathcal{I}$$

□

1.3.4 Matrix Representation of Vectors

We can express the inner product of wavefunctions more simply when the wavefunctions themselves are expressed in terms of a basis set,

$$\langle v|u\rangle = \sum_n v_n^* u_n$$

or equivalently in matrix notation,

$$\langle v|u\rangle = [v_1^*, v_2^*, \dots, v_n^*] \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}.$$

It is obvious that this is true, however, let us go through the specific details.

Proof. Suppose there are two functions, $u(\mathbf{r}) = \sum_n u_n \psi_n(\mathbf{r})$ and $v(\mathbf{r}) = \sum_m v_m \psi_m(\mathbf{r})$. Their inner product is

$$\langle v|u \rangle = \int v^*(\mathbf{r})u(\mathbf{r}) \, d\mathbf{r} = \int \left[\sum_n v_n \psi_n(\mathbf{r}) \right]^* \sum_m u_m \psi_m(\mathbf{r}) \, d\mathbf{r}.$$

Pulling out the sums,

$$\sum_n \sum_m v_n^* u_m \int \psi_n(\mathbf{r}) \psi_m(\mathbf{r}) \, d\mathbf{r} = \sum_n \sum_m v_n^* u_m \delta_{nm}.$$

However, all off-diagonals of δ_{nm} are zero, assuming that the functions are orthonormal, and so

$$\sum_{n=m} v_n^* u_n = \langle v|u \rangle$$

as before. In Dirac notation, the functions are expressed as $|u\rangle = \sum_m u_m |m\rangle$ and $|v\rangle = \sum_n v_n |n\rangle$. Noting that $\langle v| = \sum_n \langle n|v_n^*$, we get

$$\langle v|u \rangle = \sum_{n,m} \langle n|v_n^* u_m |m\rangle = \sum_{n,m} v_n^* u_m \langle n|m\rangle = \sum_{n,m} v_n^* u_m \delta_{nm}.$$

Again, the off-diagonals of δ_{nm} are zero, and so it follows that

$$\langle v|u \rangle = \sum_n v_n^* u_n$$

as before. □

With respect to a basis set, the bras and kets are identified with row and column matrices, seen above.

1.3.5 Matrix Representation of Linear Operators

Suppose we have an arbitrary linear operator, $\Omega : \mathbf{v} \rightarrow \mathbf{v}'$. It acts on a vector, $\mathbf{v}$, to map its components to construct another vector, $\mathbf{v}'$. We represent this in Dirac notation as

$$\boxed{\Omega |v\rangle = |v'\rangle.} \tag{8}$$

Linear operators adhere to the following definition

$$\boxed{\Omega[\lambda_1 |v_1\rangle + \lambda_2 |v_2\rangle] = \lambda_1 \Omega |v_1\rangle + \lambda_2 \Omega |v_2\rangle.} \tag{9}$$

With respect to an orthonormal basis set, $\{|n\rangle\}$, we can rewrite (8) as

$$\sum_m \Omega_{nm} v_m = v'_n \equiv \begin{bmatrix} \Omega_{11} & \Omega_{12} & \cdots & \Omega_{1n} \\ \Omega_{21} & \Omega_{22} & \cdots & \Omega_{2n} \\ \vdots & & \ddots & \vdots \\ \Omega_{n1} & \Omega_{n2} & \cdots & \Omega_{nn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} v'_1 \\ v'_2 \\ \vdots \\ v'_n \end{bmatrix}$$

where

$$\Omega_{nm} = \langle n|\Omega|m\rangle = \int \psi_n^*(\mathbf{r}) \Omega \psi_m(\mathbf{r}) \, d\mathbf{r}.$$

Proof. With respect to a basis set, $\{|n\rangle\}$, we can write the operator, Ω , acting on a ket, $|v\rangle$, as

$$\Omega|v\rangle = |v'\rangle \implies \Omega \sum_n v_n |n\rangle = \sum_n v'_n |n\rangle$$

where we have used (4) and its complex conjugate. Then,

$$\langle m | \sum_n v_n \Omega |n\rangle = \boxed{\langle m | \sum_n v'_n |n\rangle \equiv \int \psi_m^*(\mathbf{r}) \sum_n v_n \Omega \psi_n(\mathbf{r}) d\mathbf{r}}$$

and

$$\sum_n v_n \langle m | \Omega |n\rangle = \sum_n v'_n \langle m |n\rangle = \boxed{\sum_n v'_n \delta_{mn} = \int \psi_m^*(\mathbf{r}) \sum_n v'_n \psi_n(\mathbf{r}) d\mathbf{r}}$$

which all imply that

$$\sum_n v_n \Omega_{nm} = v'_n$$

as before. □

Borrowing from linear algebra, we say that the adjoint of a matrix, M , is the transpose and complex conjugate of itself, written

$$M_{nm}^\dagger = M_{nm}^* \quad (10)$$

We extend this to the operators, in which the adjoint of an operator, $\Omega^\dagger$, means that the operator, Ω , is associated with the bra rather than the ket. Particularly,

$$\boxed{\langle n | \Omega^\dagger |m\rangle \equiv \langle \Omega n |m\rangle} \quad (11)$$

Proof. Suppose there exists an operator, Ω , which acts on the inner product, $\langle n |m\rangle$. We can see that

$$\Omega_{nm} = \langle n | \Omega |m\rangle.$$

Recalling Ω can be represented in matrix form, by (10), we can see that

$$\Omega_{nm}^\dagger = \Omega_{nm}^* = \langle m | \Omega |n\rangle = \left[\int \psi_m^*(\mathbf{r}) \Omega \psi_n(\mathbf{r}) d\mathbf{r} \right]^* = \int [\Omega \psi_n(\mathbf{r})]^* \psi_m(\mathbf{r}) d\mathbf{r} = \langle \Omega n |m\rangle \equiv \langle n | \Omega^\dagger |m\rangle$$

which shows that

$$\int \psi_n^*(\mathbf{r}) \Omega^\dagger \psi_m(\mathbf{r}) d\mathbf{r} = \langle n | \Omega^\dagger |m\rangle$$

and

$$\int [\Omega \psi_n(\mathbf{r})]^* \psi_m(\mathbf{r}) d\mathbf{r} = \langle \Omega n |m\rangle.$$

Notice,

$$\Omega_{nm}^\dagger \equiv \langle n | \Omega^\dagger |m\rangle = \langle n | \Omega |m\rangle^\dagger.$$

□

We now state the properties of these operators without proof.

1. The adjoint of a product is $(\Omega\Lambda)^\dagger = \Lambda^\dagger\Omega^\dagger$

2. If $\Omega^\dagger = \Omega$, then Ω is *self-adjoint*. We refer to these self-adjoint operators as **Hermitian**. This relation is given by (9).

Hermitian operators are particularly important since they are related to **observables** in quantum mechanics. Their expectation values always return real quantities.

1.4 The Eigenvalue Problem

The eigenvalue problem is always of the form

$$\boxed{\Omega |\omega\rangle = w |\omega\rangle}. \quad (12)$$

Here, Ω is an operator acting on a vector (eigenvector), ω , which returns a scalar multiple of the eigenvector, with multiple w . There are two very distinct properties of a Hermitian operator.

1.4.1 Eigenvalues of a Hermitian Operator

The eigenvalues of a Hermitian operator are always real.

Proof. Let us consider (12). Taking the bra of ω on both sides,

$$\langle \omega | \Omega | \omega \rangle = w \langle \omega | \omega \rangle$$

and the adjoint of this,

$$\langle \omega | \Omega^\dagger | \omega \rangle = w^* \langle \omega | \omega \rangle^*.$$

However, if Ω is Hermitian, then

$$\langle \omega | \Omega | \omega \rangle = w^* \langle \omega | \omega \rangle \implies \langle \omega | w | \omega \rangle = w^* \langle \omega | \omega \rangle \implies w \langle \omega | \omega \rangle = w^* \langle \omega | \omega \rangle \implies w = w^*$$

for $\omega \in \mathbb{R}$. □

1.4.2 Orthogonality of Eigenfunctions

The eigenfunctions of a Hermitian operator are always orthogonal.

Proof. Suppose we have two eigenfunctions, α and β with uncommon eigenvalues. It follows that

$$\Omega |\alpha\rangle = a |\alpha\rangle \quad (1.P3)$$

$$\Omega |\beta\rangle = b |\beta\rangle \quad (1.P4)$$

for some $a, b \in \mathbb{R}$. Taking the ket of β on (1.P3) and the ket of α on (1.P4),

$$\langle \beta | \Omega | \alpha \rangle = a \langle \beta | \alpha \rangle, \quad \langle \alpha | \Omega | \beta \rangle = b \langle \alpha | \beta \rangle.$$

Subtracting the first equation from the complex conjugate of the second,

$$\langle \beta | \Omega | \alpha \rangle - \langle \alpha | \Omega | \beta \rangle^* = a \langle \beta | \alpha \rangle - b^* \langle \alpha | \beta \rangle.$$

Now, if Ω is Hermitian, then

$$\langle \beta | \Omega | \alpha \rangle - \langle \beta | \Omega | \alpha \rangle = a \langle \beta | \alpha \rangle - b^* \langle \beta | \alpha \rangle \implies 0 = (a - b^*) \langle \beta | \alpha \rangle.$$

Assuming that $a - b^* \neq 0$ (which they cannot, since we impose that the eigenfunctions are uncommon eigenvalues), for the equation to be zero, the only possible result is

$$\langle \beta | \alpha \rangle = 0.$$

□

1.5 Generalization to Continuous Basis Sets

Recall the definition for a discrete, orthonormal basis set, $\{|n\rangle\}$. We can expand this definition to a **continuous**, orthonormal basis set, represented by $\{\omega_\alpha(\mathbf{r})\}$, $\omega_\alpha(\mathbf{r})$ are functions with a continuous index, α . We thus have

$$\langle \alpha | \beta \rangle = \int \omega_\alpha^*(\mathbf{r}) \omega_\beta(\mathbf{r}) d\mathbf{r} = \delta(\alpha - \beta) \text{ (Kronecker Delta).} \quad (13)$$

$$\int |\alpha\rangle \langle \alpha| d\alpha = \mathcal{I} \text{ (Identity Operator)} \quad (14)$$

$$\int \omega_\alpha^*(\mathbf{r}') \omega_\alpha(\mathbf{r}) d\alpha = \delta(\mathbf{r} - \mathbf{r}') \text{ (Dirac delta).} \quad (15)$$

Notice that, interestingly, the Kronecker delta relation in (13) is of similar form as the Dirac delta. In fact, replacing $\beta \rightarrow \alpha$ and shifting the vector in α to $\mathbf{r} \rightarrow \mathbf{r}'$ gives (15). Thus, (13) is a generalization of (15), such that both $\omega_\alpha(\mathbf{r})$ and $\omega_\beta(\mathbf{r})$ are orthogonal, and there exists some $\omega_\beta(\mathbf{r})$ such that $\omega_\beta(\mathbf{r}) = \omega_\alpha(\mathbf{r})$.

1.5.1 Continuous Components of a Wavefunction

We can expand our wavefunctions in terms of the continuous basis sets.

$$\psi(\mathbf{r}) = \int c(\alpha) \omega_\alpha(\mathbf{r}) d\alpha, \text{ where } c(\alpha) = \int \omega_\alpha^*(\mathbf{r}) \psi(\mathbf{r}) d\mathbf{r}. \quad (16)$$

Equivalently, in Dirac notation,

$$|\psi\rangle = \int c(\alpha) |\alpha\rangle d\alpha, \text{ where } c(\alpha) = \langle \alpha | \psi \rangle$$

Proof. We begin by utilising the sifting property of the Dirac delta,

$$\psi(\mathbf{r}) = \int \psi(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') d\mathbf{r}.$$

Substituting (15) into the above equation,

$$\psi(\mathbf{r}) = \int \psi(\mathbf{r}') d\mathbf{r}' \int \omega_\alpha^*(\mathbf{r}') \omega_\alpha(\mathbf{r}) d\alpha = \int \left[\int \omega_\alpha^*(\mathbf{r}') \psi(\mathbf{r}') d\mathbf{r}' \right] \omega_\alpha(\mathbf{r}) d\alpha.$$

Comparing this to the original equation for $\psi(\mathbf{r})$, we notice that for this statement to be true, it must follow that

$$c(\alpha) \equiv \int \omega_\alpha^*(\mathbf{r}') \psi(\mathbf{r}') d\mathbf{r}' = \int \omega_\alpha^*(\mathbf{r}) \psi(\mathbf{r}) d\mathbf{r}.$$

In Dirac notation, this is represented as

$$|\psi\rangle = \mathcal{I} |\psi\rangle = \int |\alpha\rangle \langle\alpha|\psi\rangle d\alpha \implies |\psi\rangle = \int \langle\alpha|\psi\rangle |\alpha\rangle d\alpha$$

□

1.5.2 Scalar Product and Norm in Continuous Components

Let us now assume that $\psi_1(\mathbf{r})$ and $\psi_2(\mathbf{r})$ have continuous components, $b(\alpha)$ and $c(\alpha)$, each with respect to the continuous basis set, $\{|\alpha\rangle\}$. Their inner product is

$$\langle\psi_1|\psi_2\rangle = \int b^*(\alpha) c(\alpha) d\alpha$$

with a norm of

$$\langle\psi|\psi\rangle = \int |c(\alpha)|^2 d\alpha.$$

Proof. We represent both $\psi_1(\mathbf{r})$ and $\psi_2(\mathbf{r})$ in integral form,

$$\begin{aligned} \psi_1(\mathbf{r}) &= \int b(\alpha) \omega_\alpha(\mathbf{r}) d\alpha \\ \psi_2(\mathbf{r}) &= \int c(\beta) \omega_\beta(\mathbf{r}) d\beta. \end{aligned}$$

Taking the inner product and substituting,

$$\begin{aligned} \langle\psi_1|\psi_2\rangle &= \int \psi_1^*(\mathbf{r}) \psi_2(\mathbf{r}) d\mathbf{r} = \int \left[\int b^*(\alpha) \omega_\alpha^*(\mathbf{r}) d\alpha \right] \left[\int c(\beta) \omega_\beta(\mathbf{r}) d\beta \right] d\mathbf{r} \\ &= \int d\alpha \int b^*(\alpha) c(\beta) d\beta \int \omega_\alpha^*(\mathbf{r}) \omega_\beta(\mathbf{r}) d\mathbf{r}. \end{aligned}$$

Recognizing that $\int \omega_\alpha^*(\mathbf{r}) \omega_\beta(\mathbf{r}) d\mathbf{r} = \langle\alpha|\beta\rangle = \delta(\alpha - \beta)$, we arrive at

$$\langle\psi_1|\psi_2\rangle = \int d\alpha \int b^*(\alpha) c(\beta) \delta(\alpha - \beta) d\beta = \int b^*(\alpha) c(\alpha) d\alpha.$$

In Dirac notation,

$$|\psi_1\rangle = \int b(\alpha) |\alpha\rangle d\alpha \implies \langle\psi_1| = \int b^*(\alpha) \langle\alpha| d\alpha, \quad |\psi_2\rangle = \int c(\beta) |\beta\rangle d\beta.$$

Then, the inner product,

$$\langle\psi_1|\psi_2\rangle = \int d\beta \int b^*(\alpha) c(\beta) \langle\alpha|\beta\rangle d\alpha = \int b^*(\alpha) c(\alpha) d\alpha$$

as before. □

1.5.3 The Position Representation

We introduce yet another continuous basis set, $\xi_{\mathbf{r}'}(\mathbf{r})$ defined as

$$\xi_{\mathbf{r}'}(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}').$$

Here, the index, $\mathbf{r}'$, is continuous and also a vector, in that $\mathbf{r}' = x'\hat{e}_x + y'\hat{e}_y + z'\hat{e}_z$. In Dirac notation, this becomes

$$\xi_{\mathbf{r}'}(\mathbf{r}) = |\mathbf{r}'\rangle.$$

Let us confirm whether or not this in fact meets the criteria of a basis set.

Kronecker Delta. In the continuous case, we have

$$\xi_{\mathbf{r}'}(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}')$$

implies that $\xi_{\mathbf{r}'}(\mathbf{r})$ is orthonormal. We must also confirm the integral form,

$$\int \xi_{\mathbf{r}'}(\mathbf{r}) \xi_{\mathbf{r}''}(\mathbf{r}) d\mathbf{r} = \int \delta(\mathbf{r} - \mathbf{r}') \delta(\mathbf{r} - \mathbf{r}'') d\mathbf{r} = \delta(\mathbf{r}' - \mathbf{r}'') = \langle \mathbf{r}' | \mathbf{r}'' \rangle$$

which is consistent.

Identity Operator. The integral form for the Dirac delta shows that

$$\int \xi_{\mathbf{r}_0}(\mathbf{r}') \xi_{\mathbf{r}_0}(\mathbf{r}) d\mathbf{r}_0 = \int \delta(\mathbf{r}' - \mathbf{r}_0) \delta(\mathbf{r} - \mathbf{r}_0) d\mathbf{r}_0 = \delta(\mathbf{r} - \mathbf{r}')$$

which implies that $\xi_{\mathbf{r}'}(\mathbf{r})$ spans $\mathbb{H}$. Confirming the Dirac notation form,

$$\int \langle \mathbf{r}_0 | \mathbf{r}' \rangle \langle \mathbf{r} | \mathbf{r}_0 \rangle d\mathbf{r} = \delta(\mathbf{r} - \mathbf{r}')$$

which is consistent. We then conclude that $\xi_{\mathbf{r}'}(\mathbf{r})$ is valid to be a continuous basis set.

Using this position basis set, we can see that any wave vector, $|\psi\rangle$, represented with respect to $\xi_{\mathbf{r}'}(\mathbf{r})$ is written as

$$\langle \mathbf{r} | \psi \rangle = \int \xi_{\mathbf{r}}^*(\mathbf{r}') \psi(\mathbf{r}') d\mathbf{r}' = \int \delta(\mathbf{r}' - \mathbf{r}) \psi(\mathbf{r}') d\mathbf{r}' = \psi(\mathbf{r})$$

and so we come full circle to the original Dirac notation representation of the Hilbert space inner product.

$$\langle \psi_1 | \psi_2 \rangle = \langle \psi_1 | \mathcal{I} | \psi_2 \rangle = \int \langle \psi_1 | \mathbf{r} \rangle \langle \mathbf{r} | \psi_2 \rangle d\mathbf{r} = \int \psi_1^*(\mathbf{r}) \psi_2(\mathbf{r}) d\mathbf{r} \implies \langle \psi_1 | \psi_2 \rangle \equiv \int \psi_1^*(\mathbf{r}) \psi_2(\mathbf{r}) d\mathbf{r}.$$

where we have used that $\mathcal{I} = \int |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r}$.

1.5.4 The Momentum Representation

Again, we define another continuous basis set for the momentum representation of a particle,

$$v_{\mathbf{p}'}(\mathbf{r}) = (2\pi\hbar)^{-\frac{3}{2}} e^{\frac{i}{\hbar}\mathbf{p}'\cdot\mathbf{r}}$$

where $\mathbf{p}'$ is a continuous index and vector, such that $\mathbf{p}' = p'_x\hat{e}_x + p'_y\hat{e}_y + p'_z\hat{e}_z$. Abbreviating this by Dirac notation,

$$v_{\mathbf{p}'}(\mathbf{r}) = |\mathbf{p}'\rangle$$

we arrive again at the Dirac notation representation of the Hilbert space inner product,

$$\langle\psi_1|\psi_2\rangle \equiv \int \bar{\psi}_1^*(\mathbf{p})\bar{\psi}_2(\mathbf{p})\,d\mathbf{p}.$$

In fact, if we recall that the Fourier transform utilizes $e^{-i2\pi nx}$, we can let $n \rightarrow -3/2h$ and $x \rightarrow \mathbf{p} \cdot \mathbf{r}$ for a more general, 3-dimensional particle, we can see that

$$\langle\mathbf{p}|\psi\rangle = \int v_{\mathbf{p}}^*(\mathbf{r})\psi(\mathbf{r})\,d\mathbf{r} = \frac{1}{(2\pi\hbar)^{\frac{3}{2}}} \int \psi(\mathbf{r})e^{-\frac{i}{\hbar}\mathbf{p}\cdot\mathbf{r}}\,d\mathbf{r} = \psi(\mathbf{p})$$

where we have used that $\mathbf{p} = v_{\mathbf{p}}(\mathbf{r})$ to show that the position and momentum space are indeed related by a Fourier transform.

1.6 The Position and Momentum Operators

Now that we have defined the continuous bases for the position and momentum space, we can formally define their respective operators.

1.6.1 The Position Operator

We define the position operator, $\hat{X}$, to be the operator that, in the $\{|\mathbf{r}\rangle\}$ representation, multiplies by the index, x (eigenvalue problem).

$$\langle\mathbf{r}|\hat{X}|\psi\rangle = x\langle\mathbf{r}|\psi\rangle.$$

Condensing this further,

$$\hat{X}\psi(\mathbf{r}) = x\psi(\mathbf{r})$$

which implies that the $\hat{X}$ position operator has the position, x , of a quantum mechanical particle as its eigenvalues. Similarly, we can write the $\hat{Y}$ and $\hat{Z}$ position operators,

$$\begin{aligned}\langle\mathbf{r}|\hat{Y}|\psi\rangle &= y\langle\mathbf{r}|\psi\rangle \\ \langle\mathbf{r}|\hat{Z}|\psi\rangle &= z\langle\mathbf{r}|\psi\rangle.\end{aligned}$$

Combining these into a single vector operator, $\hat{\mathbf{R}}$, we get

$$\langle\mathbf{r}|\hat{\mathbf{R}}|\psi\rangle = \mathbf{r}\langle\mathbf{r}|\psi\rangle.$$

We now state an important property of operators. The **expectation value** of an operator is given by $\langle\hat{Q}\rangle$, for some arbitrary operator, $\hat{Q}$. In the case of $\hat{X}$ position operator, we can represent its expectation value as

$$\langle\hat{X}\rangle = \int \psi^*(\mathbf{r})x\psi(\mathbf{r})\,d\mathbf{r}.$$

Proof. Recall that the expectation value of a random variable is given by

$$\langle x \rangle = \int x \rho(x) dx.$$

For the case of a wavefunction, the probability density, $\rho(x)$, is simply the magnitude of the wavefunction itself. Then,

$$\langle x \rangle = \int x |\psi(x)|^2 dx.$$

Extending this to the $\hat{X}$ position operator case,

$$\langle \hat{X} \rangle = \int \psi^*(\mathbf{r}) \hat{X} \psi(\mathbf{r}) d\mathbf{r} = \int \psi^*(\mathbf{r}) x \psi(\mathbf{r}) d\mathbf{r}.$$

□

1.6.2 The Momentum Operator

Let us now define the momentum operator, $\hat{P}_x$, such that in the $\{|\mathbf{p}\rangle\}$ representation, multiplies by the index, p_x .

$$\langle \mathbf{p} | \hat{P}_x | \psi \rangle = p_x \langle \mathbf{p} | \psi \rangle.$$

The $\hat{P}_y$ and $\hat{P}_z$ momentum operators follow,

$$\langle \mathbf{p} | \hat{P}_y | \psi \rangle = p_y \langle \mathbf{p} | \psi \rangle$$

$$\langle \mathbf{p} | \hat{P}_z | \psi \rangle = p_z \langle \mathbf{p} | \psi \rangle$$

and combining into a single vector operator, $\hat{\mathbf{P}}$.

$$\langle \mathbf{p} | \hat{\mathbf{P}} | \psi \rangle = \mathbf{p} \langle \mathbf{p} | \psi \rangle.$$

Now, as in classical mechanics, $\mathbf{p}$ is related to $\mathbf{r}$ by $\vec{\nabla}$, except that there is now an additional proportionality constant $\hbar/i$. Then,

$$\langle \mathbf{r} | \hat{\mathbf{P}} | \psi \rangle = \frac{\hbar}{i} \vec{\nabla} \langle \mathbf{r} | \psi \rangle.$$

In particular, the transformation is given by

$$\hat{P}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{P}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{P}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

and thus we define the expectation value, $\langle \hat{P}_q \rangle$ for $q = x, y$ or z as

$$\langle \hat{P}_q \rangle = \int \psi^*(\mathbf{r}) \frac{\hbar}{i} \frac{\partial \psi(\mathbf{r})}{\partial q} d\mathbf{r}$$

Proof. Let us take the derivative of $\langle \hat{\mathbf{R}} \rangle$.

$$\frac{d\langle \hat{\mathbf{R}} \rangle}{dt} = \int \frac{\partial}{\partial t} \psi^*(\mathbf{r}) \mathbf{r} \psi(\mathbf{r}) d\mathbf{r} = \int \mathbf{r} \left[\psi^*(\mathbf{r}) \frac{d\psi(\mathbf{r})}{dt} - \frac{d\psi^*(\mathbf{r})}{dt} \psi(\mathbf{r}) \right] d\mathbf{r}.$$

The time derivatives of the wave functions is given by the time-independent Schrödinger equation³,

$$\begin{aligned}\frac{\partial\psi(\mathbf{r})}{\partial t} &= \frac{i\hbar}{2m}\nabla^2\psi(\mathbf{r}) - \frac{i}{\hbar}V(\mathbf{r})\psi(\mathbf{r}) \\ \frac{\partial\psi^*(\mathbf{r})}{\partial t} &= -\frac{i\hbar}{2m}\nabla^2\psi(\mathbf{r}) + \frac{i}{\hbar}V(\mathbf{r})\psi(\mathbf{r})\end{aligned}$$

such that

$$\begin{aligned}\frac{d\langle\hat{\mathbf{R}}\rangle}{dt} &= \frac{i\hbar}{2m} \int [\nabla^2\psi^*(\mathbf{r})\mathbf{r}\psi(\mathbf{r}) - \psi^*(\mathbf{r})\mathbf{r}\nabla^2\psi(\mathbf{r})] d\mathbf{r} \\ &= \frac{i\hbar}{2m} \int [\psi^*(\mathbf{r})\nabla\psi(\mathbf{r}) - \nabla\psi^*(\mathbf{r})\psi(\mathbf{r})] d\mathbf{r} \\ &= \frac{1}{m} \int \psi^*(\mathbf{r})[-i\hbar\nabla]\psi(\mathbf{r}) d\mathbf{r} \\ &= \frac{\langle\hat{\mathbf{P}}\rangle}{m}\end{aligned}$$

and thus

$$m\frac{d\langle\hat{\mathbf{R}}\rangle}{dt} = \langle\hat{\mathbf{P}}\rangle = \int \psi^*(\mathbf{r}) \left[\frac{\hbar}{i} \frac{\partial}{\partial \mathbf{q}} \right] \psi(\mathbf{r}) d\mathbf{r}$$

for $q = x, y$ or z . □

Notice the form of $\langle\hat{\mathbf{P}}\rangle = m d\langle\hat{\mathbf{R}}\rangle / dt$, which is analogous to classical mechanics, $\mathbf{p} = m\mathbf{v}$.⁵ In particular, the derivation shows that

$$\frac{d\langle\hat{\mathbf{R}}\rangle}{dt} = \langle\hat{\mathbf{v}}\rangle$$

which is the expectation value of the velocity of a particle.

1.6.3 R and P Commutation Relations and the Uncertainty Principle

We define a **commutator** of two operators, $\hat{A}$ and $\hat{B}$ to be

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \quad (17)$$

We say that two operators **commute**, if $\hat{A}\hat{B} = \hat{B}\hat{A}$, and furthermore their commutator, (17), is zero. For the operators, $\hat{R}$ and $\hat{P}$, their commutator is

$$[\hat{R}_i, \hat{P}_j] = i\hbar\delta_{ij}$$

for $i, j = x, y$ or z .

³I am not sure on whether this still holds for the time-dependent Schrödinger equation.

⁴The algebra is very messy and involves the product rule with a Laplacian. I actually had ChatGPT do this for me, but you must apply $\nabla[\mathbf{r}\psi(\mathbf{r})] = \psi\nabla\mathbf{r} + \mathbf{r}\nabla\psi(\mathbf{r})$ after using the Laplacian identity $\nabla^2 = \nabla \cdot \nabla$. From there it's just annoying simplification.

⁵This relation is called Ehrenfest's Theorem.

Proof. This proof is done in Dirac notation.⁶ Let us consider the case in x . Operating on $\psi(\mathbf{r})$ with the commutator, $[\hat{X}, \hat{P}_x]$,

$$\langle \mathbf{r} | [\hat{X}, \hat{P}_x] | \psi \rangle = \langle \mathbf{r} | \hat{X} \hat{P}_x - \hat{P}_x \hat{X} | \psi \rangle = \langle \mathbf{r} | \hat{X} \hat{P}_x | \psi \rangle - \langle \mathbf{r} | \hat{P}_x \hat{X} | \psi \rangle.$$

Notice that $\hat{P}_x | \psi \rangle$ transforms ψ , such that $\hat{P}_x | \psi \rangle : \psi \rightarrow \psi'$ and similarly $\hat{X} | \psi \rangle : \psi \rightarrow \psi''$. Substituting this back into the equation,

$$\langle \mathbf{r} | \hat{X} | \psi' \rangle - \langle \mathbf{r} | \hat{P}_x | \psi'' \rangle = x \langle \mathbf{r} | \hat{P}_x | \psi \rangle - \frac{\hbar}{i} \frac{d}{dx} (x \langle \mathbf{r} | \psi \rangle) = x \frac{\hbar}{i} \frac{d}{dx} (\langle \mathbf{r} | \psi \rangle) - \frac{\hbar}{i} \frac{d}{dx} (x \langle \mathbf{r} | \psi \rangle)$$

where we have used the relations $\hat{X} | \psi \rangle = x \hat{P}_x | \psi \rangle$ and $\hat{P}_x | \psi'' \rangle = (\hbar/i)(d/dx)x | \psi \rangle$. The second relation utilizes the implication that $\hat{X} | \psi \rangle = x | \psi \rangle$. Simplifying further,

$$\frac{\hbar}{i} \left[x \frac{d}{dx} \langle \mathbf{r} | \psi \rangle - x \frac{d}{dx} \langle \mathbf{r} | \psi \rangle - \langle \mathbf{r} | \psi \rangle \right] = i\hbar \langle \mathbf{r} | \psi \rangle = \langle \mathbf{r} | i\hbar \mathcal{I} | \psi \rangle.$$

Therefore,

$$[\hat{X}, \hat{P}_x] = i\hbar \mathcal{I}$$

and this result is the same for $\hat{Y}$ and $\hat{Z}$, with $\hat{X} \rightarrow \hat{Y}$ or $\hat{Z}$ and $x \rightarrow y$ or z . $\square$

From this commutation relation, we are able to derive the Heisenberg Uncertainty Principle, which states that

$$\Delta p_x \Delta x \geq \frac{\hbar}{2}. \quad (18)$$

and similarly has $\Delta p_y \Delta y, \Delta p_z \Delta z \geq \hbar/2$.

Proof. Let us suppose that there exists an operator $(\hat{X} + i\lambda \hat{P}_x)$ which transforms $\psi(\mathbf{r})$ such that $(\hat{X} + i\lambda \hat{P}_x) : \psi \rightarrow \psi'$. The inner product of ψ' shows that

$$\begin{aligned} \langle \psi' | \psi' \rangle &= \langle \psi | (\hat{X} - i\lambda \hat{P}_x)(\hat{X} + i\lambda \hat{P}_x) | \psi \rangle \\ &= \langle \psi | \hat{X}^2 | \psi \rangle + \langle \psi | i\lambda \hat{X} \hat{P}_x - i\lambda \hat{P}_x \hat{X} | \psi \rangle + \lambda^2 \langle \psi | \hat{P}_x^2 | \psi \rangle \\ &= \langle \hat{X}^2 \rangle + \lambda \langle [\hat{X}, \hat{P}_x] \rangle + \lambda^2 \langle \hat{P}_x^2 \rangle \\ &= \langle \hat{X}^2 \rangle - \lambda \hbar + \lambda^2 \langle \hat{P}_x^2 \rangle. \end{aligned}$$

By the Cauchy-Schwarz inequality, $\langle \psi | \psi \rangle \geq 0$ for all $x \in \mathbb{R}$, and so

$$\lambda^2 \langle \hat{P}_x^2 \rangle - \lambda \hbar + \langle \hat{X}^2 \rangle \geq 0$$

which is a quadratic in λ . For λ to admit an imaginary quantity, the discriminant of λ must also be less than or equal to zero. Thus,

$$\hbar^2 - 4 \langle \hat{P}_x^2 \rangle \langle \hat{X}^2 \rangle \leq 0 \implies \frac{\hbar^2}{4} \leq \langle \hat{P}_x^2 \rangle \langle \hat{X}^2 \rangle \implies \sqrt{\langle \hat{P}_x^2 \rangle \langle \hat{X}^2 \rangle} \geq \frac{\hbar}{2}$$

which is precisely (18) for $\Delta p = \sqrt{\langle \hat{P}_x^2 \rangle}$ and $\Delta x = \sqrt{\langle \hat{X}^2 \rangle}$. $\square$

⁶It could be done in integral form, but it necessitates the use of integration by parts (several times).

2 Postulates of Quantum Mechanics

Now that we have formally introduced the language of quantum mechanics, we are able to go through the postulates which act as the foundation of quantum mechanics. There are seven of them, as quoted from *Quantum Mechanics* by Branden and Joachain.

Postulate 1. *To an ensemble of physical systems one can, in certain cases, associate a wave function or state function which contains all the information that can be known about the ensemble. This function is in general complex; it may be multiplied by an arbitrary complex number without altering its physical significance.*

Postulate 1 originates from the wave-particle duality as observed during experiments.

Postulate 2. *The Principle of Superposition.*

$$\psi(\mathbf{r}, t) = c_1\psi_1(\mathbf{r}, t) + c_2\psi_2(\mathbf{r}, t) + \cdots + c_n\psi_n(\mathbf{r}, t) = \sum_{n=1}^{\infty} c_n\psi_n(\mathbf{r}, t). \quad (19)$$

Postulate 2 originates from the interference patterns as observed during experiments.

Proof. The Schödinger equation states

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}, t) + V(\mathbf{r}, t)\psi(\mathbf{r}, t) = \mathcal{H}\psi(\mathbf{r}, t)$$

which is a parabolic partial differential equation. In particular, the operator, $\mathcal{H}$ is a linear operator, and thus distributes itself linearly to a summation of terms. We consider the 1D case (can be generalized).

Suppose we have two wavefunctions, $\psi_1(x) = |\psi_1(x)|e^{i\alpha_1}$ and $\psi_2(x) = |\psi_2(x)|e^{i\alpha_2}$. The probability of their superpositions is

$$\begin{aligned} P(x) &= |c_1\psi_1(x) + c_2\psi_2(x)|^2 = |c_1|\psi_1(x)|e^{i\alpha_1} + c_2|\psi_2(x)|e^{i\alpha_2}|^2 \\ &= (c_1|\psi_1(x)|e^{i\alpha_1} + c_2|\psi_2(x)|e^{i\alpha_2})^* (c_1|\psi_1(x)|e^{i\alpha_1} + c_2|\psi_2(x)|e^{i\alpha_2}) \\ &= (c_1^*|\psi_1(x)|e^{-i\alpha_1} + c_2^*|\psi_2(x)|e^{-i\alpha_2}) (c_1|\psi_1(x)|e^{i\alpha_1} + c_2|\psi_2(x)|e^{i\alpha_2}) \\ &= c_1c_1^*|\psi_1(x)|^2 + c_2c_2^*|\psi_2(x)|^2 + c_1c_2^*|\psi_1(x)||\psi_2(x)|e^{i(\alpha_1-\alpha_2)} + c_2c_1^*|\psi_1(x)||\psi_2(x)|e^{i(\alpha_2-\alpha_1)} \\ &= |c_1\psi_1(x)|^2 + |c_2\psi_2(x)|^2 + c_1c_2^*|\psi_1(x)||\psi_2(x)|e^{i(\alpha_1-\alpha_2)} + \left[c_1c_2^*|\psi_1(x)||\psi_2(x)|e^{i(\alpha_1-\alpha_2)} \right]^*. \end{aligned}$$

Making use of the relation $A + A^* = 2\text{Re}(A)$,

$$P(x) = |c_1\psi_1(x)|^2 + |c_2\psi_2(x)|^2 + 2\text{Re} \left[c_1c_2^*|\psi_1(x)||\psi_2(x)|e^{i(\alpha_1-\alpha_2)} \right]$$

and notice that the quantity under the $\text{Re}(A)$ term is an interference term, which itself is another solution since $\text{Re}(c_1c_2^*) = c_3$ and $|\psi_1(x)||\psi_2(x)|e^{i(\alpha_1-\alpha_2)} = |\psi_3(x)|e^{i\alpha_3}$ for $\alpha_3 = \alpha_1 - \alpha_2$. $\square$

Postulate 3. *With every dynamical variable is associated a linear operator.*

Here, a dynamical variable is measurable, and thereby refers to an observable. However, we must take care to note that **a measurement changes the wavefunction**. Formally, we say that the wavefunction "collapses" upon being observed, and phrased mathematically, is stated as

$$\psi'(\mathbf{r}, t) = \hat{A}\psi(\mathbf{r}, t)$$

for some operator, $\hat{A}$.

Postulate 4. *The only result of a precise measurement of the dynamical variable, $\mathcal{A}$, is one of the eigenvalues, a_n , of the linear operator, $\hat{A}$ associated with $\mathcal{A}$.*

We say that the set of eigenvalues of $\hat{A}$ is the **spectrum** of $\hat{A}$. The spectrum of $\hat{A}$ can be either discrete or continuous, and those that represent a dynamical variable must be real. There are Hermitian operators.

Postulate 5. *If a series of measurements is made of a dynamical variable, $\mathcal{A}$, on an ensemble of systems described by the wavefunction, $\psi(\mathbf{r})$, the expectation value, or average value of this dynamical variable is:*

$$\langle \hat{A} \rangle = \frac{\langle \psi | \hat{A} | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\int \psi^*(\mathbf{r}) \hat{A} \psi(\mathbf{r}) d\mathbf{r}}{\int \psi^*(\mathbf{r}) \psi(\mathbf{r}) d\mathbf{r}}. \quad (20)$$

Let us extend Postulate 5 to arbitrary operators. Consider the operator, $\hat{A}$. The expectation value is

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle = \int \langle \psi | \alpha \rangle \langle \alpha | \hat{A} | \psi \rangle d\alpha.$$

We assume that $\langle \alpha | \hat{A} | \psi \rangle = a(\alpha) \langle \alpha | \psi \rangle$, and so

$$\langle \hat{A} \rangle = \int \langle \psi | \alpha \rangle a(\alpha) \langle \alpha | \psi \rangle d\alpha = \int c^*(\alpha) a(\alpha) c(\alpha) d\alpha = \int a(\alpha) |c(\alpha)|^2 d\alpha.$$

Comparing this result with (20), you may wonder why we are missing the denominating term. However, recall that we have asserted the assumption $\int |\psi(\mathbf{r})|^2 d\mathbf{r} = 1$, and so $\langle \psi | \psi \rangle = 1$.

Postulate 6. *A wavefunction representing any dynamical state can be expressed as a linear combination of the eigenfunctions of $\hat{A}$, where $\hat{A}$ is the operator associated with a dynamical variable.*

This is, in essence, the Identity Operator (7). In fact,

$$\int |\alpha \rangle \langle \alpha | d\alpha = \mathcal{I}$$

creates the set of eigenfunctions of $\hat{A}$. This can be seen in the extension from Postulate 5, but also follows from the fact that all eigenfunctions of $\hat{A}$ are orthogonal and thereby linearly independent. In principle, you return back to (19) by this argument.

Postulate 7. *The time evolution of the wavefunction of a system is determined by the time dependent Schrödinger equation:*

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \mathcal{H} \psi(\mathbf{r}, t)$$

where $\mathcal{H}$ is the Hamiltonian, or total energy operator, of the system.

3 The Schrödinger Equation

■ The generalized Schrödinger Equation (SE) is given by

$$\boxed{\mathcal{H}\psi(\mathbf{r}, t) = E\psi(\mathbf{r}, t), \text{ where } \mathcal{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t).} \quad (21)$$

and $\mathcal{H}$ is the Hamiltonian operator. In the case that $V(\mathbf{r}, t) = V(\mathbf{r})$, the SE reduces to the time-independent SE (TISE), with a general solution made of a linear combination of stationary states. We show how this is derived later in chapter 3.2. In 1D, specifically, the wavefunction is governed by

$$\psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-i\frac{E_n}{\hbar}t}$$

and considering only the x -dependence term, the TISE becomes

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_n(x) + V(x)\psi_n(x) = E_n \psi_n(x).} \quad (22)$$

3.1 Particle in a Box

Let us now consider a one-dimensional potential well, governed by the potential

$$V(x) = \begin{cases} 0, & x \in [0, a] \\ \infty, & \text{else.} \end{cases} \quad (23)$$

The potential demands that outside the box, the wave function must be zero, while inside the box, the wavefunction is some eigenfunction of the TISE. Solving (22) with $V(x) = 0 \forall x \in [0, a]$,

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \cancel{V(x)\psi(x)} = E\psi(x) \implies \frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2} \psi(x).$$

Notice that this is in the form of simple harmonic motion, with $k^2 = 2mE/\hbar^2$, then

$$\psi(x) = A \cos(kx) + B \sin(kx).$$

The boundary conditions of the problem demand that $\psi(0) = \psi(a) = 0$, and so

$$\begin{aligned} \psi(0) &\implies 0 = A \cos(0) + \cancel{B \sin(0)} = A \implies A = 0 \\ \psi(a) &\implies 0 = \cancel{A \cos(ka)} + B \sin(ka) \implies B \sin(ka) = 0. \end{aligned}$$

For the second relation to be true, it must be that

$$ka = n\pi \implies k = \frac{n\pi}{a} = \frac{\sqrt{2mE}}{\hbar} \implies E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}.$$

Furthermore, applying the normalization condition that $\int |\psi(x)|^2 dx = 1$,

$$1 = \int_0^a |\psi(x)|^2 dx = B^2 \int_0^a \sin^2\left(\frac{n\pi}{a}x\right) dx = B^2 \left[\frac{x}{2} - \frac{a \sin(2n\pi x/a)}{4n\pi} \right]_0^a = B^2 \frac{a}{2} \implies B = \sqrt{\frac{2}{a}}.$$

We arrive at the solution,

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right), \quad E_n = \frac{n^2\pi^2\hbar^2}{2ma^2} \quad (24)$$

where E_n are the eigenstates (eigenvalues) and $\psi_n(x)$ are the eigenfunctions of the TISE.

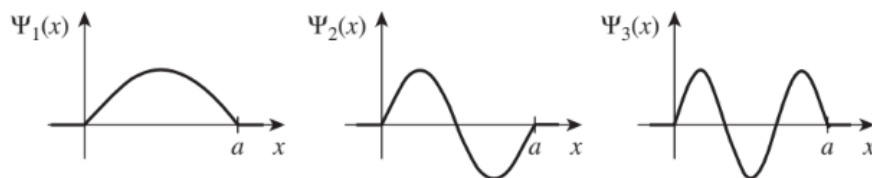


Figure 2: First three eigenstates of the particle in a box.

Notice, in the limiting case that $E_n \rightarrow \infty$, $\psi_n(x)$ becomes a free particle, since E_n converges (becomes continuous) in a spectrum.

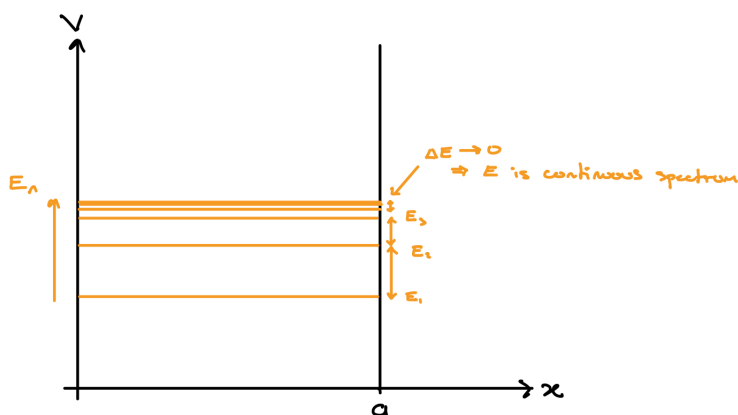


Figure 3: Limiting case that $E_n \rightarrow \infty$. The eigenstates converge and become continuous.



There are a few important properties that correspond to the x -dependent solutions of the TISE. These properties are a consequence of the most general type of equation of the x -dependence for the TISE, called a **Sturm-Liouville Equation**⁷. Beginning with the first,

1. All eigenstates of the TISE are real, with eigenfunctions that may not necessarily be real.
2. All eigenstates of the TISE form an infinite sequence and can be ordered in increasing magnitude such that

$$E_1 < E_2 < \cdots < E_n \rightarrow \infty$$

⁷I add additional properties that are not shown in Griffith's that I think are important to know, taken from other courses, notably AMATH 351.

as $n \rightarrow \infty$.

3. Each solution, $\psi_n(x)$, has $(n - 1)$ node within $x \in [0, a]$.
4. Each solution is mutually orthogonal, which is simply the orthogonality property in Chapter 1.3.1.
5. The solutions, $\psi_n(x)$, are even and odd with respect to the center of the well, such that all $n = (2p + 1)$ solutions are even, and all $n = 2p$ solutions are odd, for $p \in \mathbb{Z}$.

3.2 Stationary States

Chapter 3.1 shows that the solution to the TISE is only a function that depends on x , $\psi(x)$. However, we know that the wavefunction itself has both an spatial, x , and time, t , dependence.  Let us show how this is derived. Generally, the TISE is a partial differential equation, so we will use this representation of (21). For this case, the SE is given by

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x)\psi(x, t)$$

and here is the important part: *we look for solutions that are products of its components.* That is, we use the method of **separation of variables**. So, we have

$$\psi(x, t) = \psi(x)\phi(t)$$

where $\psi(x)$ is really $\psi_n(x)$ and is what we have solved for in the particle in a box, and $\phi(t)$ is some function of t alone. If we take the partial derivatives (which become ordinary) for the SE assuming our new form,

$$\frac{\partial \psi(x, t)}{\partial t} = \psi(x) \frac{d\phi(t)}{dt}, \quad \frac{\partial^2 \psi(x, t)}{\partial x^2} = \phi(t) \frac{d^2 \psi(x)}{dx^2}$$

and substituting it back in again, we get

$$i\hbar \psi(x) \frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m} \phi(t) \frac{d^2 \psi(x)}{dx^2} + V(x)\psi(x)\phi(t).$$

Dividing through by $\psi(x)\phi(t)$, we can rearrange this into

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x).$$

Notice that this new form separates the variables, and for this to be valid, it must be that both sides are equal to some constant. In particular, we cannot vary the variable, t , while expecting x to compensate. Furthermore, you might notice that we have already *solved* for the right side, and so let us deal with the left. As our solution for $\psi_n(x)$ may imply, we let the solution in $\phi(t)$ be equal to the constant, E_n , and this gives

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = E_n \implies \frac{1}{\phi(t)} d\phi(t) = \frac{E_n}{i\hbar} dt \implies \ln |\phi(t)| = \frac{E_n}{i\hbar} t$$

giving the solution, after adding the subscript n to help distinguish between the different states of $\phi(t)$,

$$\phi_n(t) = e^{-i \frac{E_n}{\hbar} t}.$$

■ Thus, using our solution in $\psi(x)$, we can finally write the full, generalized solution.

$$\psi(x, t) = \psi(x)\phi(t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{\frac{-iE_n t}{\hbar}} \implies \boxed{\psi(x, t) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi}{a}x\right) e^{-i\frac{n^2\pi^2\hbar}{2ma^2}t}} \quad (25)$$

where we have added the sum since by Postulate 6, the wavefunction is actually a linear combination of its eigenfunctions. Using Dirac notation, we can find the coefficients, c_n , by considering only $\psi(x)$,

$$\begin{aligned} |\psi\rangle &= \sum_n c_n |n\rangle \implies |\psi(0)\rangle = \sum_n c_n |n\rangle \\ \langle m|\psi(0)\rangle &= \sum_n c_n \langle m|n\rangle = \delta_{nm} = c_m \end{aligned}$$

where we have used

$$\langle x|\psi(0)\rangle = \sum_n c_n \langle x|n\rangle \implies \psi(x, 0) = \sum_n c_n \psi_n(x)$$

and

$$c_n = \langle n|\psi(0)\rangle = \int \langle n|x\rangle \langle x|\psi(0)\rangle dx = \int \psi_n^*(x) \psi(x, 0) dx.$$

Thus, we can see that $\psi_n^*(x) = \psi_n(x)$ (it is real), and so the coefficients are given by

$$\boxed{c_n = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right) \psi(x, 0) dx} \quad (26)$$

■ Here, also, there are some things to note about the stationary state solutions of $\psi(x, t)$.

1. Each independent solution, $\psi_n(x, t)$, is a **stationary state**. In particular, its probability density is independent of time, and so

$$|\psi_n(x, t)|^2 = \psi_n^*(x, t) \psi_n(x, t) = |\psi_n(x)|^2$$

and every expectation value of $\psi_n(x, t)$ is also constant in time.⁸

2. All eigenstates are states of *definite total energy*, which means that every measurement of $\mathcal{H}$ will return the value, E . We see this by the variance of $\mathcal{H}$,

$$\sigma_{\mathcal{H}}^2 = \langle \mathcal{H}^2 \rangle - \langle \mathcal{H} \rangle^2 = E^2 - E^2 = 0.$$

3. The stationary state solution set, (25), converges to $[\psi(x+, t) + \psi(x-, t)]/2$ at each point on $x \in (0, a)$, where $\psi(x\pm)$ refer to the positively and negatively-sided limits of $\psi(x, t)$.
4. The solution set, (25), is complete, such that any $\psi_m(x, t)$ can be expressed as a linear combination of other existing solutions, $c_n \psi_n(x, t)$.

⁸It's worth being careful here. The general solution in (25) doesn't have time independence in the probability density, since for different stationary states, the energies are also different.

You might also make the connection that (25) is simply a **Fourier series** for $\psi(x, t)$, and that the expansion property in 4 is given by **Dirichlet's Theorem**.

Example 3.1. Suppose we are given the wavefunction $\psi(x, 0) = Ax(a - x)$, suspended in a infinite potential well of length, a . Normalizing the wavefunction,

$$\begin{aligned} 1 &= A^2 \int_0^a x^2(a - x)^2 dx = A^2 \int_0^a x^2(a^2 - 2ax + x^2) dx \\ &= A^2 \left[a^2 \int_0^a x^2 dx - 2a \int_0^a x^3 dx + \int_0^a x^4 dx \right] \\ &= A^2 \left[a^2 \frac{a^3}{3} - 2a \frac{a^4}{4} + \frac{a^5}{5} \right] \end{aligned}$$

which shows that

$$\frac{A^3 a^5}{30} = 1 \implies A = \sqrt{\frac{30}{a^5}}.$$

We find the coefficients by (26),

$$\begin{aligned} c_n &= \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right) \psi(x, 0) dx = \sqrt{\frac{60}{a^6}} \left[\int_0^a \sin\left(\frac{n\pi}{a}x\right) x(a - x) dx \right] \\ &= \sqrt{\frac{60}{a^6}} \left[\int_0^a x \sin\left(\frac{n\pi}{a}x\right) dx - \int_0^a x^2 \sin\left(\frac{n\pi}{a}x\right) dx \right] \\ &= \sqrt{\frac{60}{a^6}} [I_1 + I_2]. \end{aligned}$$

By integration by parts,

$$\int_a^b f \frac{dg}{dx} dx = fg \Big|_a^b - \int_a^b \frac{df}{dx} g dx$$

shows that

$$\begin{aligned} I_1 &= -\frac{a}{n\pi} x \cos\left(\frac{n\pi}{a}x\right) \Big|_0^a + \frac{a}{n\pi} \int_0^a \cos\left(\frac{n\pi}{a}x\right) dx = -\frac{a^2}{n\pi} \cos(n\pi) \implies I_1 = -\frac{a^2}{n\pi} (-1)^n \\ I_2 &= -\frac{ax^2}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \Big|_0^a + \frac{2a}{n\pi} \int_0^a x \cos\left(\frac{n\pi}{a}x\right) dx \\ &= -\frac{ax^2}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \Big|_0^a + \frac{2a}{n\pi} \left[x \frac{a}{n\pi} \sin\left(\frac{n\pi}{a}x\right) \Big|_0^a - \int_0^a \frac{a}{n\pi} \sin\left(\frac{n\pi}{a}x\right) dx \right] \\ &= -\frac{ax^2}{n\pi} \cos\left(\frac{n\pi}{a}x\right) \Big|_0^a + \left[\frac{2a}{n\pi} \right]^2 [(-1)^n - 1] \\ &= -\frac{a^3}{n\pi} (-1)^n + 4 \left[\frac{a^3}{n\pi} \right]^3 [(-1)^n - 1] \end{aligned}$$

So,

$$c_n = -2\sqrt{\frac{60}{a^6}} \left[\frac{a}{n\pi} \right]^3 [(-1)^n - 1] = \begin{cases} \frac{8\sqrt{15}}{(n\pi)^3}, & \text{if } n \text{ is odd.} \\ 0, & \text{if } n \text{ is even.} \end{cases}$$

Then, the wavefunction becomes

$$\psi(x, t) = \sqrt{\frac{30}{a} \left[\frac{2}{\pi} \right]^3} \sum_{p=1}^{\infty} \left\{ \left[\frac{\sin[(2p+1)\pi x a^{-1}]}{(2p+1)^3} \right] e^{-\frac{i(2p+1)^2 \pi^2 \hbar}{2ma^2} t} \right\}$$

where we have let $p = 2n + 1$ to assimilate all odd integers of n into all the reals.

3.3 The Harmonic Oscillator

■ We know the harmonic oscillator is represented by the force equation, $F = -kx$, with a potential energy, $V = kx^2/2$. Many potentials reduce to the form of V when x is small, and so we can use this as a first approximation to understand many phenomena as harmonic oscillations.

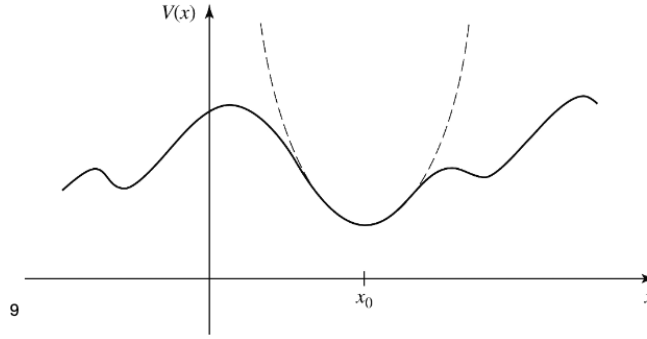


Figure 4: First approximation of a potential in the form $V = kx^2/2$, centered at x_0 .

Formally, we Taylor expand about the point x_0 ,

$$V(x) = \cancel{V(x_0)} + \frac{dV(x_0)}{dx}(x - x_0) + \frac{1}{2} \frac{d^2V(x_0)}{dx^2}(x - x_0)^2 + \dots$$

and as long as $(x - x_0)$ is small, terms higher than second order are negligible. For the quantum picture, we solve the Schrödinger equation for the potential

$$V(X) = \frac{1}{2} m \omega^2 \hat{X}^2, \quad \text{where } \omega = \sqrt{\frac{k}{m}}.$$

3.3.1 Wavefunctions and Energies

Substituting the new potential, the time-independent Schrödinger equation becomes

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2} m \omega^2 \hat{X}^2 \psi(x) = E \psi(x).} \quad (27)$$

Notice that this has an infinite number of discrete solutions. The ground state (lowest energy) of the quantum harmonic oscillator is then

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x^2}, \quad E_0 = \frac{1}{2} \hbar \omega.$$

Higher energy solutions correspond to

$$\psi_n(x) = A_n(a_+)^n \psi_0(x), \quad E_n = \left(n + \frac{1}{2}\right) \hbar \omega$$

where we have introduced a_+ as the **raising operator** to find all the higher energy wave functions and their energies. The related **lowering operator**, a_- , and a_+ together are called ladder operators. We attempt to rerepresent that Hamiltonian using ladder operators, as below. First, let us write the TISE in Dirac notation as in the 1D case as follows,

$$\langle x | \mathcal{H} | \psi \rangle = \langle x | E | \psi \rangle \implies \mathcal{H} \psi(x) = E \psi(x).$$

Recalling (27), and using the momentum operator, $\hat{P} = -i\hbar d/dx$,

$$\frac{1}{2m} \left[\hat{P}^2 + (m\omega \hat{X})^2 \right] \psi(x) = E \psi(x).$$

Now, we make the odd (but it works) step that we ignore the use of the operators, treating them as just regular variables. Now,

$$\hat{a}_{\pm} = \frac{1}{\sqrt{2\hbar m\omega}} (\mp i\hat{P} + m\omega \hat{X}) \quad (28)$$

and decomposing this into operator products,

$$\begin{aligned} \hat{a}_+ \hat{a}_- &= \frac{1}{2\hbar m\omega} (-i\hat{P} + m\omega \hat{X})(i\hat{P} + m\omega \hat{X}) \\ &= \frac{1}{2\hbar m\omega} \left[\hat{P}^2 + (m\omega \hat{X})^2 + im\omega (\hat{X}\hat{P} - \hat{P}\hat{X}) \right] \\ &= \frac{1}{2\hbar m\omega} \left[\hat{P}^2 + (m\omega \hat{X})^2 \right] + \frac{i}{2\hbar} [\hat{X}, \hat{P}]. \end{aligned}$$

We have already worked out the commutator $[\hat{X}, \hat{P}]$, and you may recognize that $[\hat{P}^2 + (m\omega \hat{X})^2]/m = \mathcal{H}$. Then,

$$\hat{a}_+ \hat{a}_- = \frac{1}{\hbar \omega} \mathcal{H} - \frac{1}{2} \implies \mathcal{H} = \hbar \omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

which is just an alternate representation of the Hamiltonian in (22). In particular, this result for the Hamiltonian has almost the exact same form as the eigenstates for the wavefunction. By inspection, we can see that $\hat{a}_+ \hat{a}_- = n$, and $\mathcal{H} = E_n$. Additionally, we can also see from the form of (28) has a nice symmetry, and we may wonder if the commutator of $\hat{a}_{\pm}$ gives a nice result (hint: it does). Working this out specifically, we can see that

$$[\hat{a}_-, \hat{a}_+] = \hat{a}_- \hat{a}_+ - \hat{a}_+ \hat{a}_- = \frac{1}{\hbar \omega} \mathcal{H} + \frac{1}{2} - \frac{1}{\hbar \omega} \mathcal{H} + \frac{1}{2} = 1.$$

Let us use this result to see how we can solve the TISE by ladder operators.

Example 3.2. The TISE can be expanded to give

$$\begin{aligned} \mathcal{H} \hat{a}_+ \psi(x) &= (E + \hbar \omega) \hat{a}_+ \psi(x) \\ &= \hbar \omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) \hat{a}_+ \psi(x) \\ &= \hbar \omega \left(\hat{a}_+ \hat{a}_- \hat{a}_+ + \frac{1}{2} \hat{a}_+ \right) \psi(x) \\ &= \hbar \omega \hat{a}_+ \left(\hat{a}_- \hat{a}_+ + \frac{1}{2} + 1 \right) \psi(x). \end{aligned}$$

There is an important note here. We had to go through the effort of distributing $\hat{a}_+$ into the parentheses, then pulling it out again to flip the signs of $\hat{a}_+\hat{a}_- \rightarrow \hat{a}_-\hat{a}_+$. We also pick up an extra +1 factor from this process. Generally, you **cannot** just bring $\hat{a}_+$ to the other side (or any operator in particular).

$$\begin{aligned}\mathcal{H}\hat{a}_+\psi(x) &= \hat{a}_+\left[\hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2} + 1\right)\psi(x)\right] \\ &= \hat{a}_+\left[\hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right) + \hbar\omega\right]\psi(x).\end{aligned}$$

Recalling the alternate representation of $\mathcal{H}$,

$$\begin{aligned}\mathcal{H}\hat{a}_+\psi(x) &= \hat{a}_+(\mathcal{H} + \hbar\omega)\psi(x) \\ &= \hat{a}_+[\mathcal{H}\psi(x)] \\ &= \hat{a}_+(E + \hbar\omega)\psi(x)\end{aligned}$$

and so we arrive at

$$\mathcal{H}\hat{a}_\pm\psi(x) = (E \pm \hbar\omega)\hat{a}_\pm\psi(x).$$

Example 3.3. Using the result of Ex. 3.2, find the ground state energy of the wavefunction by the use of ladder operators. We start by imposing $\hat{a}_-\psi_0(x) = 0$.⁹

$$\hat{a}_-\psi_0(x) = \frac{1}{\sqrt{2\hbar m\omega}}\left(\hbar\frac{d}{dx} + m\omega\hat{X}\right)\psi_0(x) = 0.$$

Here, we have used the relation $\hat{P} = (\hbar/i)(d/dx)$. Distributing $\psi_0(x)$ into the parentheses and separating terms, we can see that

$$\frac{d\psi_0(x)}{dx} = -\frac{m\omega}{\hbar}\hat{X}\psi_0(x) = -\frac{m\omega}{\hbar}x\psi_0(x).$$

Solving the first order separable differential equation (DE),

$$\ln|\psi_0(x)| = -\frac{m\omega}{2\hbar}x^2 + C \implies \psi_0(x) = Ae^{-\frac{m\omega}{2\hbar}x^2}.$$

The normalization condition of the wavefunction demands that $\int |\psi_0(x)|^2 dx = 1$, and so

$$|A|^2 \int e^{-\frac{m\omega}{\hbar}x^2} dx = |A|^2 \sqrt{\frac{\pi\hbar}{m\omega}} = 1 \implies A^2 = \sqrt{\frac{m\omega}{\pi\hbar}}$$

thus,

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}.$$

The associated ground energy state can be found by the TISE.

$$\begin{aligned}\mathcal{H}\psi_0(x) &= E_0\psi_0(x) \\ \hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right)\psi_0(x) &= E_0\psi_0(x) \\ \cancel{\hbar\omega\hat{a}_+\hat{a}_-\psi_0(x)} + \frac{\hbar\omega}{2}\psi_0(x) &= E_0\psi_0(x) \\ \frac{\hbar\omega}{2} &= E_0.\end{aligned}$$

⁹The lowest step in the ladder you can climb is zero for the ladder operators. Physically, this corresponds to $\hat{a}_\pm = 0$ for the ground state.

Example 3.4. Use the result of Ex. 3.3 to determine the remaining energy states of the wavefunction. Using again the TISE,

$$\mathcal{H}\hat{a}_+\psi_0(x) = (E_0 + \hbar\omega)\hat{a}_+\psi_0(x)$$

where we have used $E_1 = E_0 + \hbar\omega$. Making the substitution $\hat{a}_+\psi_0(x) = \alpha\psi_1(x)$, since the raising operator increases the energy,

$$\begin{aligned}\psi_1(x) &= A_1\hat{a}_+\psi_0(x) = \frac{A_1}{\sqrt{2\hbar m\omega}} \left(-\hbar \frac{d}{dx} + m\omega\hat{X} \right) \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2} \\ &= \frac{A_1}{\sqrt{2\hbar m\omega}} \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} (m\omega x + m\omega x) e^{-\frac{m\omega}{2\hbar}x^2} \\ &= A_1 \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} \sqrt{\frac{2m\omega}{\hbar}} \left(x e^{-\frac{m\omega}{2\hbar}x^2} \right).\end{aligned}$$

Normalizing the wavefunction,

$$\begin{aligned}1 &= |A_1|^2 \sqrt{\frac{m\omega}{\pi\hbar}} \left(\frac{2m\omega}{\hbar} \right) \int_{-\infty}^{\infty} x^2 e^{-\frac{m\omega}{\hbar}x^2} dx = |A_1|^2 \sqrt{\frac{m\omega}{\pi\hbar}} \left(\frac{2m\omega}{\hbar} \right) \frac{\sqrt{\pi}}{2} \left(\frac{m\omega}{\hbar} \right)^{-\frac{3}{2}} \\ &= |A_1|^2 \cancel{\sqrt{\frac{m\omega}{\hbar}}} \cancel{\sqrt{\frac{\hbar}{m\omega}}}\end{aligned}$$

and therefore $A_1 = 1$. Thus,

$$\psi_1(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} \sqrt{\frac{2m\omega}{\hbar}} \left(x e^{-\frac{m\omega}{2\hbar}x^2} \right).$$



Examples 3.3 and 3.4 result in a very important kind of polynomial called the **Hermite polynomials**. To see this more clearly, if we let $a_0 = (m\omega/\pi\hbar)^{1/4}$ and $a_1 = a_0\sqrt{2m\omega/\hbar} = (m\omega/\pi\hbar)^{1/4}\sqrt{2m\omega/\hbar}$, we can rewrite $\psi_1(x)$ and $\psi_0(x)$ in Ex. 3.4 and 3.3 above as

$$\begin{aligned}\psi_1(x) &= [a_1x + a_0]e^{-\frac{m\omega}{2\hbar}x^2} \\ \psi_0(x) &= a_0e^{-\frac{m\omega}{2\hbar}x^2}\end{aligned}$$

where, if you solved for the higher energy eigenfunctions, you can see that a polynomial of degree n in $x\sqrt{m\omega/\hbar}$ appears as the pre-factor for $\exp(-m\omega x^2/2\hbar)$. This polynomial is given by $H_n(\xi = x\sqrt{m\omega/\hbar})$, and is specifically of the form

$$H_{n+1}(\xi) = 2\xi H_n(\xi) - 2nH_{n-1}(\xi) \quad (29)$$

which is a recurrence relation. The first few Hermite polynomials are listed below which generalize to its recurrence relation.

$$\begin{aligned}H_0(\xi) &= 1 \\ H_1(\xi) &= 2\xi \\ H_2(\xi) &= 4\xi^2 - 2 \\ H_3(\xi) &= 8\xi^3 - 12\xi \\ H_4(\xi) &= 16\xi^4 - 48\xi^2 + 12 \\ &\vdots \\ H_{n-1}(\xi) &= \frac{2\xi H_n(\xi) - H_{n+1}(\xi)}{2n}.\end{aligned} \quad \text{(Hermite Polynomials)}$$

3.3.2 The Number Operator

We define the **number operator** as $\hat{N} = \hat{a}_+ \hat{a}_-$, such that

$$\boxed{\hat{N}\psi_n(x) = \hat{a}_+ \hat{a}_- \psi_n(x) = n\psi_n(x).} \quad (30)$$

Proof. Let us expand (30).

$$\hat{a}_+ \hat{a}_- \psi_n(x) = \frac{1}{\hbar\omega} \mathcal{H}\psi_n(x) - \frac{1}{2}\psi_n(x).$$

Using the relation that $\mathcal{H}\psi_n(x) = E_n\psi_n(x)$,

$$\hat{a}_+ \hat{a}_- \psi_n(x) = \frac{1}{\hbar\omega} E_n \psi_n(x) - \frac{1}{2}\psi_n(x) = \frac{1}{\hbar\omega} \left(n + \frac{1}{2}\right) \hbar\omega \psi_n(x) - \frac{1}{2}\psi_n(x) = n\psi_n(x)$$

and we so arrive at (30). $\square$

We can also do the opposite operation, $\hat{a}_- \hat{a}_+$ on $\psi_n(x)$, which is not all too different. In fact,

$$\begin{aligned} \hat{a}_- \hat{a}_+ \psi_n(x) &= \frac{1}{\hbar\omega} \mathcal{H}\psi_n(x) + \frac{1}{2}\psi_n(x) = \frac{1}{\hbar\omega} E_n \psi_n(x) + \frac{1}{2}\psi_n(x) \\ &= \frac{1}{\hbar\omega} \left(n + \frac{1}{2}\right) \hbar\omega \psi_n(x) + \frac{1}{2}\psi_n(x) \end{aligned}$$

which simplifies to

$$\hat{a}_- \hat{a}_+ \psi_n(x) = (n+1)\psi_n(x).$$

3.3.3 Normalization

We are also able to normalize wavefunctions of the harmonic oscillator by the use of the ladder operators. To see how this is the case, we begin with the usual constraint that we impose on $\psi_n(x)$.

$$\int |\psi_n(x)|^2 dx = 1 \implies \langle n|n \rangle = 1.$$

Now, let us assume that $\psi_0(x)$ is normalized, such that

$$\langle 0|0 \rangle = 1.$$

By the ladder operators, we can find $\psi_1(x)$,

$$\psi_1(x) = A_1 \hat{a}_+ \psi_0(x) \implies |1\rangle = A_1 \hat{a}_+ |0\rangle \implies \langle 1| = A_1^* \langle 0| \hat{a}_-$$

where we have used the relationship that $\hat{a}_+^\dagger = \hat{a}_-$ and $\hat{a}_-^\dagger = \hat{a}_+$. Acting both $\langle 1|$ and $|1\rangle$ on each other,

$$\langle 1|1 \rangle = |A_1|^2 \langle 0| \hat{a}_- \hat{a}_+ |0 \rangle.$$

Now, we recall the commutator of $\hat{a}_\pm$, where in particular $[\hat{a}_-, \hat{a}_+] = 1$. This shows that $\hat{a}_- \hat{a}_+ = 1 + \hat{a}_+ \hat{a}_-$ and so,

$$\langle 1|1 \rangle = |A_1|^2 \langle 0| \hat{a}_+ \hat{a}_- + 1 |0 \rangle = |A_1|^2 \left[\langle 0| \hat{a}_+ \hat{a}_- |0 \rangle + \langle 0|0 \rangle \right] = |A_1|^2 = 1.$$

The first inner product is zero since $\hat{N} = \hat{a}_+ \hat{a}_-$ returns back the index n , which is zero, and we have already assumed that $\langle 0|0 \rangle = 1$, to show that

$$\langle 1|1 \rangle = |A_1|^2 = 1.$$

Let us now look at the case $n = 2$. In particular,

$$|2\rangle = A_2 \hat{a}_+ |1\rangle \implies \langle 2|2 \rangle = |A_2|^2 \langle 1| \hat{a}_- \hat{a}_+ |1\rangle = |A_2|^2 \langle 1| \hat{a}_+ \hat{a}_- + 1 |1\rangle = |A_2|^2 [\langle 1| \hat{a}_+ \hat{a}_- |1\rangle + \langle 1|1 \rangle].$$

Again, the number operator returns the index, so

$$\langle 2|2 \rangle = |A_2|^2 [\langle 1|1 \rangle + \langle 1|1 \rangle] = 2|A_2|^2 \implies |A_2|^2 = \frac{1}{2}.$$

since $\langle 2|2 \rangle = 1$. The $n = 3$ case translates similarly,

$$\langle 3|3 \rangle = |A_3|^2 \langle 2| \hat{a}_- \hat{a}_+ |2\rangle = |A_3|^2 [\langle 2| \hat{a}_+ \hat{a}_- |2\rangle + \langle 2|2 \rangle] \implies |A_3|^2 = \frac{1}{3}$$

We can now extend these ideas to any index, n , where in particular the jump from the ground state, $\psi_0(x)$ to $\psi_n(x)$ is given by

$$\boxed{\psi_n(x) = \frac{1}{\sqrt{n!}} (\hat{a}_+)^n \psi_0(x), \text{ where } E_n = \left(n + \frac{1}{2}\right) \hbar \omega.} \quad (31)$$

In a recurrence relation, we can represent $\psi_n(x)$ in terms of its previous iteration, $\psi_{n-1}(x)$, such that

$$\psi_n(x) = \frac{1}{\sqrt{n}} \hat{a}_+ \psi_{n-1}(x) \quad (32)$$

or alternatively, other forms are

$$\hat{a}_+ \psi_n(x) = \sqrt{n+1} \psi_{n+1}(x) \quad (33)$$

$$\hat{a}_- \psi_n(x) = \sqrt{n} \psi_{n-1}(x). \quad (34)$$

It is important to make the following distinction: *all quantum mechanical systems are defined to have an index, n , such that n ranges from 0 to 1, **except** for the ground state, $n = 0$, which ranges from 0 to n by convention.*

Example 3.5. Find the expectation value of the potential energy in the n -th state of the harmonic oscillator. Let us begin by recalling $\langle \hat{V} \rangle$, where

$$\langle \hat{V} \rangle = \left\langle \frac{1}{2} m \omega^2 \hat{X}^2 \right\rangle = \frac{1}{2} m \omega^2 \langle \hat{X}^2 \rangle.$$

Making use of the Identity Operator, $\mathcal{I}$, we can rewrite this as

$$\langle \hat{V} \rangle = \frac{1}{2} m \omega^2 \langle n | \mathcal{I} \hat{X}^2 | n \rangle = \frac{1}{2} m \omega^2 \int \langle n | x \rangle \langle x | \hat{X}^2 | n \rangle dx = \frac{1}{2} m \omega^2 \int \psi_n^*(x) \hat{X}^2 \psi_n(x) dx.$$

Now, using the ladder operators, we quickly find

$$\hat{a}_+ + \hat{a}_- = \frac{2m\omega \hat{X}}{\sqrt{2\hbar m\omega}} \implies \hat{X} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-).$$

The momentum follows,

$$\hat{a}_+ + \hat{a}_- = \frac{-2i\hat{P}}{\sqrt{2\hbar m\omega}} \implies \hat{P} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}_+ + \hat{a}_-).$$

Returning back to the original example, we substitute $\hat{X}$ into our expression for $\langle \hat{V} \rangle$, so

$$\langle \hat{V} \rangle = \frac{1}{2}m\omega^2 \int \psi_n^*(x) \left[\frac{\hbar}{2m\omega} (\hat{a}_+^2 + \hat{a}_+ \hat{a}_- + \hat{a}_- \hat{a}_+ + \hat{a}_-^2) \right] \psi_n(x) dx.$$

We now make use of the fact that some of the ladder operators will return an index that is not n . In particular, we can see that

$$\begin{aligned} \hat{a}_+^2 \psi_n(x) &\propto \psi_{n+2}(x) \\ \hat{a}_-^2 \psi_n(x) &\propto \psi_{n-2}(x). \end{aligned}$$

Defining $m = n + 2$ and $p = n - 2$, we end up with an inner product of the form $\psi_n^*(x)\psi_m(x)$ or $\psi_n^*(x)\psi_p(x)$ which correspond to $\hat{a}_+^2$ and $\hat{a}_-^2$ respectively. Thus, we can condense $\langle \hat{V} \rangle$ to

$$\begin{aligned} \langle \hat{V} \rangle &= \frac{1}{2}m\omega^2 \frac{\hbar}{2m\omega} \left[n \int \psi_n^*(x) \psi_n(x) dx + (n+1) \int \psi_n^*(x) \psi_n(x) dx \right] \\ &= \frac{\hbar\omega}{4}(2n+1). \end{aligned}$$

Factoring out 2,

$$\langle \hat{V} \rangle = \frac{1}{2}\hbar\omega \left(n + \frac{1}{2} \right).$$

We can now revisit our previous discussion on the Hermite polynomials. For an n th state wavefunction, we can write its form in terms of the Hermite polynomials as

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\frac{\xi^2}{2}}. \quad (35)$$

In this form, we can see how the quantum harmonic oscillator evolves for greater integers of n .

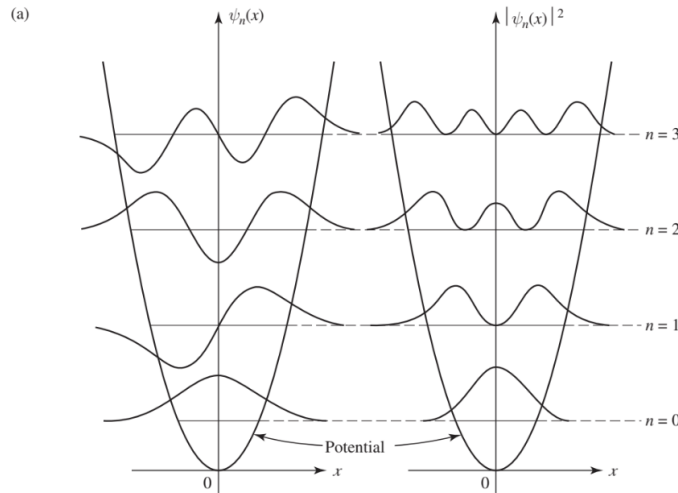


Figure 5: $n = 1$ to $n = 4$ states of the quantum harmonic oscillator.

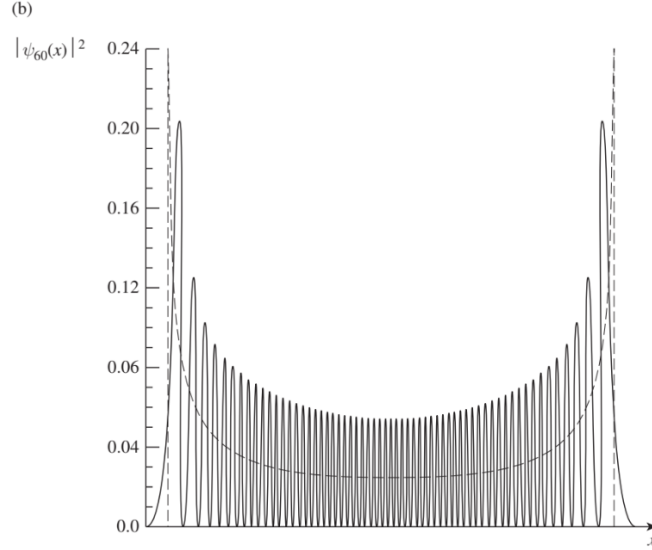


Figure 6: Graph of $|\psi_{60}|^2$ with the classical distribution (dashed curve) superimposed.

In particular, notice that the probability of finding the particle outside the classically allowed range is non-zero, and for all odd states of n , the probability of finding the particle at the center of the potential is zero. In the case that $n \rightarrow \infty$, we can see that the quantum oscillator begins to recover the classical behaviour.

3.4 The Free Particle

3.4.1 Traveling Wave Solution

A free particle in 1D obeys the TISE with the constraint $V(x) = 0$ everywhere, such that

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x).$$

Let us now derive the traveling wave solution for a free particle. For convenience, we define $k \equiv \sqrt{2mE}/\hbar$, and we solve the TISE, which has solutions of the form

$$\psi(x) = A \sin(kx) + B \cos(kx).$$

By Euler's formula¹⁰, we can rewrite $\psi(x)$ as

$$\psi(x) = A \frac{e^{ikx}}{2i} - A \frac{e^{-ikx}}{2i} + B \frac{e^{ikx}}{2} + B \frac{e^{-ikx}}{2} = \left(\frac{A}{2i} + \frac{B}{2} \right) e^{ikx} + \left(\frac{B}{2} - \frac{A}{2i} \right) e^{-ikx}.$$

We now redefine the constants, such that the pre-factors of $\exp(ikx)$ and $\exp(-ikx)$ are C and D , and so the solution becomes

$$\psi(x) = Ce^{ikx} + De^{-ikx}.$$

¹⁰Euler's identity says that $e^{ix} = \cos(x) + i\sin(x)$. In particular, $\cos(x) = \text{Re}(e^{ix}) = (e^{ix} + e^{-ix})/2$ and $\sin(x) = \text{Im}(e^{ix}) = (e^{ix} - e^{-ix})/2i$.

Given our previous derivation in Chapter 3.2, we can now write the full solution of the free particle, for the time-dependent factor $\phi(t) = \exp(-iEt/\hbar) = \exp(k^2\hbar/2m)$.

$$\psi(x, t) = \psi(x)\phi(t) = Ce^{ik(x - \frac{\hbar k}{2m}t)} + De^{-ik(x + \frac{\hbar k}{2m}t)}.$$

Or, more compactly,

$$\boxed{\psi(x, t) = A_i e^{ik(x - \frac{\hbar k}{2m}t)}, \text{ where } k \equiv \pm \frac{\sqrt{2mE}}{\hbar}.} \quad (36)$$

Notice, for $k > 0$, the motion of $\psi(x, t)$ corresponds to a wave traveling to the right, while for $k < 0$, $\psi(x, t)$ becomes is a wave traveling to the left. Here, the amplitude of $\psi(x, t)$ is given the subscript i , as we cannot expect the amplitude to be the same for both left-and-right traveling cases. We now make the important observation:

$$\left[\frac{\hbar k}{m} \right] = \frac{[J][s]}{[kg]} \frac{\sqrt{[kg][J]}}{[J][s]} = \frac{\sqrt{[kg][N][m]}}{[kg]} = \frac{\sqrt{[kg][kg][m][m][s^{-2}]}}{[kg]} = \frac{[kg][m]}{[kg][s]} = \frac{[m]}{[s]}$$

which are the units of speed.¹¹ With this idea, we can condense (36) further, where $v = \hbar k/m$, and so

$$\psi(x, t) = A_i e^{ik(x - \frac{v}{2}t)}, \text{ where } v \equiv \pm \frac{\hbar k}{m} \quad (37)$$

where now $v > 0$ corresponds to $\psi(x, t)$ traveling to the right, while for $v < 0$, $\psi(x, t)$ is a wave traveling to the left. Immediately, we can see that by our new definition of v , $\psi(x, t)$ appears to travel at half the speed of the particle it is supposed to represent. Furthermore, (36) is **not integrable**, and therefore not normalizable, implying that it **cannot** exist in Hilbert space. Let us attempt to convince ourselves otherwise. First, we recall the *classical* wave-vector, $k = 2\pi/\lambda$, and substitute this into the expression for kinetic energy,

$$T = \frac{1}{2}mv^2 = \frac{k^2\hbar^2}{2m} = \frac{(2\pi)^2\hbar^2}{2m\lambda^2} \implies \lambda = \frac{2\pi\hbar}{mv} = \frac{h}{mv} = \frac{h}{p} \quad (38)$$

where we have switched from the original reduced Planck's constant to the regular Planck constant by $\hbar = h/2\pi$. The term, $\lambda = h/p$, is known as the **de Broglie wavelength**, and is a consequence of the quantum mechanical system we are dealing with. In fact, let us reconcile the differences between the two. The two speeds we are dealing with are

$$v_{\text{classical}} = \sqrt{\frac{2E}{m}}, \quad v_{\text{quantum}} = \frac{\hbar k}{2m}.$$

In terms of the de Broglie wavelength, we can rewrite k in the classical wavelength,

$$k = \frac{2\pi}{\lambda} = 2\pi \frac{mv_{\text{classical}}}{h} = \frac{mv_{\text{classical}}}{\hbar}$$

and substituting back into the quantum rendition,

$$v_{\text{quantum}} = \frac{\hbar}{2m} \frac{mv_{\text{classical}}}{\hbar} \implies \frac{v_{\text{quantum}}}{v_{\text{classical}}} = \frac{1}{2}$$

¹¹Notice the classical equivalent. In classical wave theory, $\psi(x, t) \propto \exp(x - vt)$, where v is the speed of the wave. In fact, $v = \omega/k$ in classical theory, which is exactly the same as in the quantum counterpart; the only difference to the classical counterpart, is that k is not $\sqrt{2mE}/\hbar$.

which shows that the quantum speed of the wavefunction is *half* of what it would be in the classical interpretation. We now discuss the non-integrability of the free particle. Beginning with the assumption on the quantum systems,

$$\int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = |A|^2 \int_{-\infty}^{\infty} e^{-ik(x - \frac{\hbar k}{2m}t)} e^{ik(x - \frac{\hbar k}{2m}t)} dx = |A|^2 \lim_{l \rightarrow \infty} \lim_{s \rightarrow -\infty} \left[x \right]_s^l = |A|^2 [l - s]$$

which is clearly divergent for s and t . The solution to these problems is to treat the wave number, k , as a continuous index and form *new* solution using a superposition of all k values. In fact, our previous definition of k made a subtle assumption on the value of k , in that it is *strictly* a single value for all (x, t) . In reality, this is not the case and leads to an incomplete solution (wrong, actually) entirely. Let us now derive the proper solution. Beginning with $\psi(x, t)$ as before,

$$\psi(x, t) = A e^{ik(x - \frac{\hbar k}{2m}t)}$$

and we now sum over all k 's to reconstruct $\psi(x, t)$ as a function of a continuous index, k .

$$\Psi(x, t) = \sum_k A_k e^{ik(x - \frac{\hbar k}{2m}t)}.$$

Now, attempting to normalize this function again, we have

$$\Psi(x, t) = \int_{-\infty}^{\infty} \left[\sum_k A_k e^{ik(x - \frac{\hbar k}{2m}t)} \right]^2 dx \implies \boxed{\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ik(x - \frac{\hbar k}{2m}t)} dk.} \quad (39)$$

where $\phi(k) \sim A_k$, taking its place, and a pre-factor of $1/\sqrt{2\pi}$ is pulled out after simplification. Furthermore, let us assume $t = 0$ and determine the form of the integral of $\Psi(x, 0)$.

$$\boxed{\Psi(x, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx} dk.} \quad (40)$$

From this form, we can see that (40) is an **inverse Fourier transform** of the function, $\phi(k)$. Naturally, the **Fourier transform** arises in symmetric fashion,¹²

$$\boxed{\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx.} \quad (41)$$

Example 3.6. Consider a free particle, initially localized in the range $x \in [-a, a]$, and is released at time $t = 0$.

$$\Psi(x, 0) = \begin{cases} A, & x \in [-a, a] \\ 0, & \text{else.} \end{cases}$$

where A and a are positive, real constants. Find $\Psi(x, t)$. We begin by normalizing the wavefunction,

$$\int_{-\infty}^{\infty} |\Psi(x, 0)|^2 dx = |A|^2 \int_{-a}^a dx = |A|^2 (2a) = 1 \implies A = \frac{1}{\sqrt{2a}}.$$

¹²This is actually a consequence of Plancherel's Theorem, which states that $\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$, where $\hat{f}(\xi)$ is the frequency spectrum of $f(x)$. In particular, Plancherel's theorem demands that the functions $\Psi(x, 0)$ and $\phi(k)$ exist and are therefore normalizable.

Defining $\phi(k)$, we use (41),

$$\begin{aligned}\phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx = \frac{1}{2\sqrt{a\pi}} \int_{-a}^a e^{-ikx} dx \\ &= \frac{1}{2\sqrt{a\pi}} \left[\frac{e^{-ikx}}{-ik} \right]_{-a}^a \\ &= \frac{1}{m\sqrt{a\pi}} \left[\frac{e^{ika} - e^{-ika}}{2i} \right].\end{aligned}$$

By Euler's formula,

$$\phi(k) = \sqrt{\frac{a}{\pi}} \frac{\sin(ka)}{ka} = \sqrt{\frac{a}{\pi}} \text{sinc}(ka).$$

We reconstruct, $\Psi(x, t)$ using (40),

$$\Psi(x, t) = \frac{1}{\pi} \sqrt{\frac{a}{2}} \int_{-\infty}^{\infty} \text{sinc}(ka) e^{ik(x - \frac{\hbar k}{2m}t)} dk$$

which is not analytically solvable, but numerically integrable.

3.4.2 Phase and Group Velocities

Let us return to the classical and quantum velocities that we found earlier. In particular, we denote the quantum velocity as the **phase velocity** of $\psi(x, t)$, while the classical velocity is the **group velocity**. The wave-packet that is constructed from these two velocities are a superposition of sinusoidal functions with amplitudes modulated by $\phi(k)$.

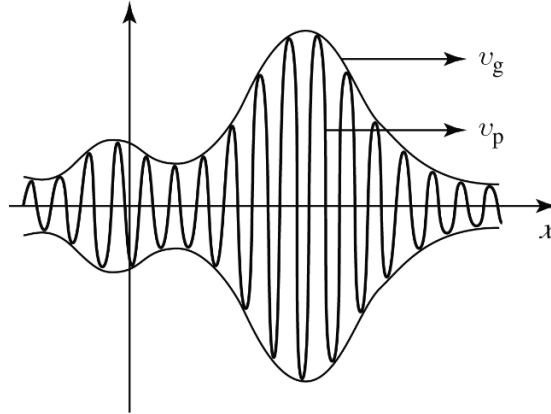


Figure 7: Wave-packet of $\psi(x, t)$ as modulated by $\phi(k)$ and constructed by v_g and v_p .

Notice, from Fig. 7, the wave-packet consists of small "ripples" within an "envelope". Furthermore, the speed of these smaller waves is the phase velocity, while the speed of the entire envelope is the group velocity. For the free particle, the group velocity is twice the speed of the phase velocity. Let us show this is the case. First, we use (40).

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ik(x - \frac{\hbar k}{2m}t)} dk.$$

Let us now define a new quantity, called the **dispersion relation**, such that $\omega(k) = \hbar k^2/2m$.¹³ Then, we find that

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \omega t)} dk.$$

Now, expanding $\omega(k)$ in a Taylor series centered about $k = k_0$, we have

$$\omega(k) \approx \omega(k_0) + \left. \frac{d\omega}{dk} \right|_{k_0} (k - k_0)$$

for higher order terms removed. Making the new substitution, $s = k - k_0 \implies ds = dk$, we can see that

$$\begin{aligned} \Psi(x, t) &\approx \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(s + k_0) e^{i\{[s+k_0]x - [\omega(k_0) + \left. \frac{d\omega}{dk} \right|_{k_0} s]t\}} ds \\ &\approx \frac{1}{\sqrt{2\pi}} e^{i[k_0 x - \omega(k_0)t]} \int_{-\infty}^{\infty} \phi(s + k_0) e^{is\left[x - \left. \frac{d\omega}{dk} \right|_{k_0} t\right]} ds. \end{aligned}$$

Condensing $\Psi(x, t)$ further, we let $\omega(k_0) = \omega_0$ and $d\omega/dk|_{k_0} = \omega'_0$, giving

$$\boxed{\Psi(x, t) = \frac{1}{\sqrt{2\pi}} e^{i(k_0 x - \omega_0 t)} \int_{-\infty}^{\infty} \phi(k_0 + s) e^{is(x - \omega'_0 t)} ds.} \quad (42)$$

Notice, (42) is of the form $\Psi(x, t) = A\psi_p(x, t)\psi_g(x, t)$, where in particular,

$$\begin{aligned} \psi_p(x, t) &= e^{i(k_0 x - \omega_0 t)} \\ \psi_g(x, t) &= \int_{-\infty}^{\infty} \phi(k_0 + s) e^{is(x - \omega'_0 t)} ds \end{aligned}$$

and the indices p and g refer to the components of $\Psi(x, t)$ governed by the phase and group velocity of $\Psi(x, t)$. We can see that $\psi_p(x, t)$ is responsible for the "ripples"¹⁴ contained within $\Psi(x, t)$, while $\psi_g(x, t)$ is responsible for the "envelope" which contains it. In this way, we say that $\psi_g(x, t)$ **modulates** $\psi_p(x, t)$. Furthermore, notice that $\psi_p(x, t)$ travels at a speed (constant),

$$v_p = \frac{\omega_0}{k_0}$$

while $\psi_g(x, t)$ travels at a speed,

$$v_g = \left. \frac{d\omega}{dk} \right|_{k_0}.$$

We now must confirm if this holds with our previous result, that $v_p/v_g = 1/2$. Dropping the subscript 0 since it was already arbitrary and recalling that $\omega = \hbar k^2/2m$, we find that

$$v_p = \frac{\omega}{k} = \frac{\hbar k}{2m}, \quad v_g = \frac{d\omega}{dk} = \frac{\hbar k}{m}.$$

Then, their ratios are

$$\frac{v_p}{v_g} = \frac{\hbar k}{2m} \frac{m}{\hbar k} = \frac{1}{2}$$

as before, confirming our result. In particular, $v_{\text{classical}} = v_g = 2v_p$.

¹³ $\omega(k)$ is not always $\hbar k^2/2m$. In fact, for an arbitrary wave-packet, the form of $\omega(k)$ may differ.

¹⁴In a sense, it's obvious this is true because we assume that the wave packet has a constant pre-factor, $\exp[i(k_0 x - \omega_0 t)]$, governed by the first term in the Taylor expansion of ω . For more complicated systems, this may not be the case.

3.5 The Delta Function Potential

3.5.1 Bound and Scattering States

In Chapters 3.1-3.3, we discussed the infinite square well and the harmonic oscillators. They had normalizable states with discrete indices. These properties are *unique*, and we refer to them as **bound states**. On the other hand, in Chapter 3.4, we discussed the free particle states. These were non-normalizable and had continuous indices of k ; they are called **scattering states**.

There is an important significance to defining these two types of states, and we can understand this by making an analogy to classical mechanics. Let us consider a 1D, time-independent potential, $V = V(x)$ and a particle within the bounds $x \in [a, b]$.

Bound States. Consider the case $V(x) > E$, where $E = T + U$, such that E is the total energy of a particle. In this case, the particle is confined within a potential well, that is, $E < V(x) \forall x \in [a, b] \cup [c, d]$, and there is no physical way for the converse inequality to be true. The particle itself may oscillate between the **turning points** of $V(x)$, but it is never able to escape without being provided an extra source of energy. This is what we observe in the infinite potential well.

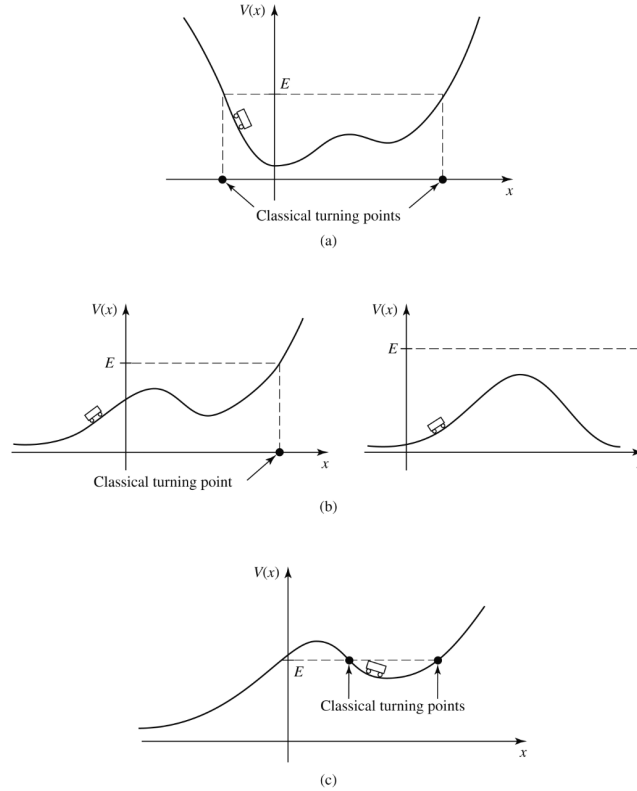


Figure 8: (a) Bound state of a particle, where $E < V(x)$ within $x \in (-\infty, a] \cup [b, \infty)$. (b) Left: scattering state of a particle, where $E > V(x)$ within $x \in (-\infty, c)$ but $E < V(x)$ within $x \in (c, \infty)$. Right: scattering state where $E > V(x)$ for $x \in (-\infty, \infty)$. (c) Classical bound state, but a quantum scattering state, where $E < V(x)$ within $x \in [a, b] \cup [c, \infty)$, but $E > V(x)$, for $x \in (-\infty, d] \cup [e, f]$.

Scattering States. Now, suppose $E > V(x)$, where again $E = T + U$, that is, $E > V(x) \forall x \in [a, b] \cup [c, d]$ or $x \in [a, b]$ or $x \in [c, d]$. In this case, the particle enters the potential from infinity, has motion governed by $V(x)$ temporarily, then returns to infinity. Here, there is no way for the particle to become trapped in the potential, unless it loses energy to some external source. This is what we observe in the free particle.

Now, there is in fact a general theorem which captures scattering states: *a scattering state may occur if the energy of the particle, E , is such that $E > \min(V)$, provided that $V(x)$ is finite and $V(x) < E$ on at least one side of $x \rightarrow \pm\infty$.*

These two types of states correspond to the two kinds of solutions that the SE permits. In fact, our distinction between the two motivate the inequality condition for the types of states. We are interested in the potential at infinity, and so

$$\begin{cases} E < V(\pm\infty) & \implies \text{bound state} \\ E > V(-\infty) \text{ or } V(\infty) & \implies \text{scattering state.} \end{cases}$$

However, in reality, most potentials go to zero at infinity, so this condition simplifies to

$$\begin{cases} E < 0 & \implies \text{bound state} \\ E > 0 & \implies \text{scattering state.} \end{cases}$$

In the case of the infinite square well and the harmonic oscillator, the potentials which govern $\psi(x, t)$ tend to infinity as $x \rightarrow \pm\infty$, arising the consequence of bound states. For the free particle, the potential is zero everywhere, but the energy of the particle is not, fulfilling $E > 0$.

For the case of the free particle with $E \leq 0$, the general solution set of $\psi(x, t)$ is incomplete, and is thus non-normalizable for all $x \in (-\infty, \infty)$.

3.5.2 The Delta Function Well

Now, we direct our attention to a specific kind of well: the Delta-function well, which satisfies both kinds of states. The Delta-function well is governed by the potential form,

$$V(x) = -\alpha\delta(x) \tag{43}$$

for some constant α ¹⁵. This potential is very artificial, but illustrates the idea behind the different states. The TISE in this case becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} - \alpha\delta(x)\psi(x) = E\psi(x)$$

and yields both bound states for $E < 0$ and scattering states for $E > 0$. Let us consider first the bound state.

Case $E < 0$. Because of the non-zero potential of $\delta(x)$ at $x = 0$, we must separate the solutions of $E < 0$ into two regions: $x \in (-\infty, 0)$ and $x \in (0, \infty)$. In the region $x < 0$, we find that $V(x) = 0$, and so

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\psi(x) = k^2\psi(x), \quad k \equiv \frac{\sqrt{-2mE}}{\hbar}.$$

¹⁵ $\delta(x)$ carries units of inverse length, so α must have units of energy $\times$ length.

Note that the negative under the root is still valid, since $E < 0$ by assumption. The solution is of the form of simple harmonic motion (SHM), so

$$\psi(x) = Ae^{-kx} + Be^{kx}.$$

But as $x \rightarrow -\infty$, $\exp(-kx) \rightarrow \infty$, so we must discard A for $\psi(x)$ to be valid. Thus,

$$\psi(x) = Be^{kx}, \quad (x < 0).$$

In the region $x > 0$, we again have $V(x) = 0$, and the TISE is the same as before. The solution is

$$\psi(x) = Ce^{-kx} + De^{kx}$$

but as $x \rightarrow \infty$, $\exp(kx) \rightarrow \infty$, so we must discard D as before. Thus,

$$\psi(x) = De^{-kx}, \quad (x > 0).$$

It only remains to construct $\psi(x)$ as a piecewise function within the space $x \in (-\infty, 0] \cup [0, \infty)$. However, there are important boundary conditions on ψ .

1. $\psi(x)$ is continuous, such that $\psi(x) \in C^0(\mathbb{R})$.
2. $d\psi(x)/dx$ is continuous for all points of x except 0, such that $d\psi(x)/dx \in C^0(\mathbb{R}; x \neq 0)$.

From the first condition, we must have

$$\psi(0) \Big|_{x<0} = \psi(0) \Big|_{x>0} \implies B = D$$

and so we can write the full solution $\psi(x)$ as

$$\psi(x) = \begin{cases} Be^{kx}, & x \in (-\infty, 0] \\ Be^{-kx}, & x \in [0, \infty). \end{cases} \quad (44)$$

The second condition, however, does not say anything about the solution; it is the specific case where $V(x=0) = -\alpha\delta(0) = -\infty$. In fact, because we discarded the portions of the solution where $\psi(x) \rightarrow \infty$, the second condition is effectively null for $\psi(x)$ *until* we invoke its derivative.

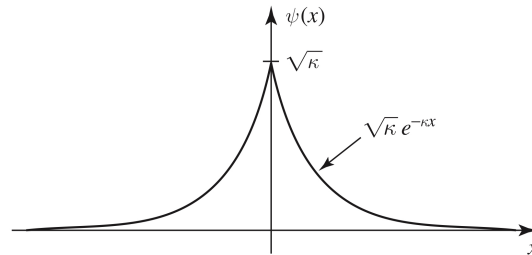


Figure 9: Piecewise solution, $\psi(x)$, to the Delta-function well along the real line.

Because we cannot just evaluate the derivative at $x = \lim_{\epsilon \rightarrow 0} \epsilon = 0$ (it is undefined), we must instead go through the effort of taking the integral Schrödinger's equation from some bounds $x \in [-\epsilon, +\epsilon]$, then take the limit as $\epsilon \rightarrow 0$. This gives,

$$-\frac{\hbar^2}{2m} \int_{-\epsilon}^{+\epsilon} \frac{d^2\psi(x)}{dx^2} dx + \int_{-\epsilon}^{+\epsilon} V(x)\psi(x) dx = E \int_{-\epsilon}^{+\epsilon} \psi(x) dx.$$

Notice that the first integral is simply the derivative of $\psi(x)$ with respect to x evaluated at both endpoints, while the last integral is zero in the limit $\epsilon \rightarrow 0$. Why? The integral from $x \in [-\epsilon, +\epsilon]$ evaluates the area of $\psi(x)$ of an infinitesimally small region, so the area is also infinitesimally small. In the limit that $\epsilon \rightarrow 0$, we have $A \rightarrow 0$, as before. If you're unconvinced (as I was), we can also show this explicitly.

Let us suppose that we Taylor expand $\psi(x)$ to the first-order. *Be careful!* We do **not** Taylor expand to the second-order *because* we know that the derivative, $d\psi(x)/dx$ is discontinuous at $x = 0$. Then, in a small neighborhood about the point $x = 0$, we have

$$\psi(x) = \psi(0) + \mathcal{O}(x).$$

We can capture this more clearly by stating that there exists a function

$$\varepsilon(x) = \mathcal{O}(x) = \psi(x) - \psi(0).$$

We note that for some sufficiently small ϵ within the boundary $[-\epsilon, +\epsilon]$, we have that

$$|\varepsilon(x)| = |\psi(x) - \psi(0)| < \delta$$

for some quantity $\delta > 0$. Then, evaluating $\varepsilon(x)$ within the given bounds, we can see that

$$\left| \int_{-\epsilon}^{+\epsilon} \varepsilon(x) dx \right| \leq \int_{-\epsilon}^{+\epsilon} |\varepsilon(x)| dx < \int_{-\epsilon}^{+\epsilon} \delta dx = 2\epsilon \delta.$$

where the first inequality is a well-known property of integrals. It is easy to see that as we evaluate $\epsilon \rightarrow 0$, then

$$\lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} \varepsilon(x) dx \leq \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} |\varepsilon(x)| dx = 0.$$

Returning back to our original Taylor expansion, we also evaluate within the bounds, to show

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} \psi(x) dx &= \lim_{\epsilon \rightarrow 0} \left[\psi(0) \int_{-\epsilon}^{+\epsilon} dx \right] + \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} \cancel{\varepsilon(x)} dx \\ &= \lim_{\epsilon \rightarrow 0} [2\epsilon\psi(0)] \\ &= 0 \end{aligned}$$

which is zero, as expected. Now, let us return to the original problem, such that

$$-\frac{\hbar^2}{2m} \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} \frac{d}{dx} \left[\frac{d\psi(x)}{dx} \right] dx + \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} V(x)\psi(x) dx = 0.$$

Rearranging, and using the fundamental theorem of calculus, we see that

$$\Delta \left[\frac{d\psi(x)}{dx} \right] \equiv \lim_{\epsilon \rightarrow 0} \left[\frac{d\psi(x)}{dx} \Big|_{+\epsilon} - \frac{d\psi(x)}{dx} \Big|_{-\epsilon} \right] = \frac{2m}{\hbar^2} \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} V(x)\psi(x) dx.$$

Unlike before, we cannot say that the right-hand side is zero, since $\psi(x)$ is now multiplied by the quantity $V(x)$. So, if $V(x) = -\alpha\delta(x)$, we find that by the sifting property for $\delta(x)$ centered at $x_0 = 0$,

$$\frac{2m}{\hbar^2} \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} V(x)\psi(x) dx = -\frac{2m}{\hbar^2} \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{+\epsilon} \alpha\delta(x)\psi(x) dx = -\frac{2m\alpha}{\hbar^2} \psi(0).$$

Thus, we arrive at

$$\Delta \left[\frac{d\psi(x)}{dx} \right] = -\frac{2m\alpha}{\hbar^2} \psi(0).$$

We now recall (44) and determine the values of $d\psi(x)/dx$ for the two regions $x > 0$ and $x < 0$. In particular, we have that $\epsilon > 0$ corresponds to $x > 0$, while $\epsilon < 0$ corresponds to $x < 0$, such that

$$\left. \frac{d\psi(x)}{dx} \right|_{+\epsilon} = -Bke^{-kx}, \quad \left. \frac{d\psi(x)}{dx} \right|_{-\epsilon} = Bke^{kx}.$$

Then, since $\epsilon \rightarrow 0 \implies x \rightarrow 0$, we have

$$\lim_{\epsilon \rightarrow 0} [-Bke^{-k\epsilon} - Bke^{k\epsilon}] = -2Bk, \quad \psi(0) = B.$$

Thus,

$$-2Bk = -\frac{2m\alpha}{\hbar^2} B \implies k = \frac{m\alpha}{\hbar^2}.$$

Our energy eigenstates then become

$$\frac{m^2\alpha^2}{\hbar^4} = -\frac{2mE}{\hbar^2} \implies E = -\frac{m\alpha^2}{2\hbar^2}.$$

Now, we are finally able to normalize $\psi(x)$, such that

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 2|B|^2 \int_0^{\infty} e^{-2kx} dx = \frac{2|B|^2}{-2k} \left[e^{-2kx} \right]_0^{\infty} = \frac{|B|^2}{k} = 1 \implies B = \sqrt{k} = \frac{\sqrt{m\alpha}}{\hbar}$$

where we have chosen the positive, real root. Thus, we can finally state our solution for the bound state, which is

$$\boxed{\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-\frac{m\alpha|x|}{\hbar^2}}, \quad E = -\frac{m\alpha^2}{2\hbar^2}} \quad (45)$$

where we have added the absolute value sign to encapsulate both the negative and positive regions of the delta function well. Notice this result also shows that the well has *exactly* one single bound state, regardless of the quantity α .

Case $E > 0$. Let us now consider the scattering states, for $E > 0$. Now, the Schrödinger equation is

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \psi(x) = -k^2 \psi(x), \quad k \equiv \frac{\sqrt{2mE}}{\hbar}$$

where k is real and positive. We again split the real line into two regions, $x < 0$ and $x > 0$. For $x < 0$, we have that the solution is of the form

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \quad (x < 0)$$

where neither term blows up due to the imaginary dependence of the exponential. For $x > 0$, we have

$$\psi(x) = Fe^{ikx} + Ge^{-ikx}, \quad (x > 0).$$

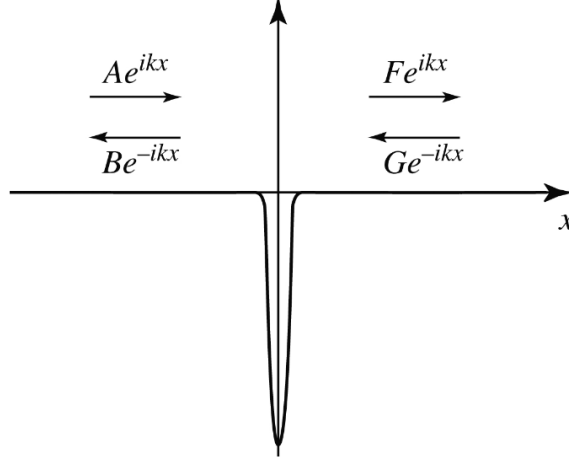


Figure 10: Solutions before validity of $\psi(x)$ along the Delta-function well. Notice that $A \exp(ikx) \rightarrow \delta(x = 0)$ and $G \exp(-ikx) \rightarrow \delta(x = 0)$.

Constructing $\psi(x)$ again as a piecewise solution, we find that

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x \in (-\infty, 0] \\ Fe^{ikx} + Ge^{-ikx}, & x \in [0, \infty). \end{cases}$$

By inspection, we see that the first boundary condition at $x = 0$ imposes that

$$F + G = A + B.$$

Repeating the same analysis as in the the case $E < 0$, we find that the derivatives are

$$\begin{aligned} \left. \frac{d\psi(x)}{dx} \right|_{x>0} &= ik(Fe^{ikx} - Ge^{-ikx}) \implies \lim_{\epsilon \rightarrow 0} \frac{d\psi(x)}{dx} = ik(F - G) \\ \left. \frac{d\psi(x)}{dx} \right|_{x<0} &= ik(Ae^{ikx} - Be^{-ikx}) \implies \lim_{\epsilon \rightarrow 0} \frac{d\psi(x)}{dx} = ik(A - B). \end{aligned}$$

Then, it follows that the second boundary condition shows that

$$\Delta \left[\frac{d\psi(x)}{dx} \right] = ik(F - G - A + B) = -\frac{2m\alpha}{\hbar^2}(A + B).$$

We can write this in a nicer form, that is

$$F - G = A(1 + 2i\beta) - B(1 - 2i\beta), \quad \text{where } \beta \equiv \frac{m\alpha}{\hbar^2 k}.$$

We are now left with two equations and four unknowns (A , B , F , G , and k). This is **not** normalizable, since we can at most determine a single constant. What, then, does this state mean?

Let us recall that $\exp(ikx)$ indicates a wave traveling to the *right*, and $\exp(-ikx)$ indicates a wave traveling to the left. If this is the case, we can re-examine our previous general solutions to see what exactly is happening.

For the solution in A and B , we can see, from Fig. 10, that A is the amplitude of a wave entering the well from the left, while B is the amplitude of a wave returning to the left. For the solution in F and G , F is the amplitude of a wave traveling to the right, while G is the amplitude of a wave entering the well from the right. In fact, we can take an alternative view to see that the two solutions in the four constants create a superposition in both regions.

Throughout the real line, the solution components $A \exp(ikx)$ and $F \exp(ikx)$ effectively create a wave traveling in all space toward the right, while the solution components $B \exp(-ikx)$ and $G \exp(-ikx)$ effectively create a wave traveling in all space toward the left. In essence, we can create an *effective* solution (which is really a rewriting of our original piecewise solution), that is

$$\psi_{\text{eff}}(x) = (A + F)e^{ikx} + (B + G)e^{-ikx}.$$

In reality, however, scattering experiments have particles being fired from a single direction, so we can simplify our original solution to depend on these initial conditions.

Let us suppose that $G = 0$ to indicate scattering from the left. Now, A is the amplitude of our **incident wave**, B is the amplitude of our **reflected wave**, while F is the amplitude of the **transmitted wave**. Our system of equations now become

$$\begin{aligned} F &= A + B \\ F &= A(1 + 2i\beta) - B(1 - 2i\beta). \end{aligned}$$

Solving for both F and B , we find that

$$B = \frac{i\beta}{1 - i\beta}A, \quad F = \frac{1}{1 - i\beta}A.$$

The probability of finding a particle at any given location is given by $|\Psi(x, t)|^2$, but since the $\psi(x)$ is not normalizable, $\Psi(x, t)$ will also not be normalizable. Thus, we instead opt for the *relative* probability, which we define as R , such that

$$R \equiv \frac{|B|^2}{|A|^2} = \frac{\beta^2}{1 + \beta^2}. \quad (46)$$

Here, we call R the **reflection coefficient**. On the other hand, the relative probability that a particle will pass straight through is

$$T \equiv \frac{|F|^2}{|A|^2} = \frac{1}{1 + \beta^2} \quad (47)$$

which we define as the **transmission coefficient**. Note that in this case, the velocity of the particle is constant on both sides of the well; in general, this is not the case, which makes the form of our probabilities slightly different.

Naturally, we expect the sum of the probabilities to be one,

$$R + T = \frac{\beta^2}{1 + \beta^2} + \frac{1}{1 + \beta^2} = \frac{\beta^2 + 1}{1 + \beta^2} = 1$$

as we expected. Now, since R and T are functions of β , and hence functions of E , we can rewrite this in terms of our eigenstates as

$$\boxed{R = \frac{1}{1 + (2\hbar^2 E / m\alpha^2)}, \quad T = \frac{1}{1 + (m\alpha^2 / 2\hbar^2 E)}}. \quad (48)$$

which implies that the higher the energy, the greater the probability of transmission.

We can translate this result to the delta function *barrier*, where rather than having $V(x) = -\alpha\delta(x)$, we now have $V(x) = \alpha\delta(x)$. Here, we cannot have bound states with $E < 0$, but it *is* possible to still have a scattering state. In fact, we call this phenomenon **tunneling**, where the particle is free to move past the barrier despite its infinite height.

To see this qualitatively, let us consider the limit $\alpha \rightarrow -\infty$. We thus have, for $E < 0$,

$$\psi(x) = -\frac{m\alpha}{\hbar} e^{\frac{m\alpha|x|}{\hbar^2}}.$$

Notice, $\psi(x)$ in this case is **not normalizable**, showing that we *cannot* have a bound state within a delta-function barrier. We can imagine this as the particle being *unable* to be trapped within the well, since... it doesn't exist. This furthermore agrees with our previous condition on E for the two types of states.

On the other hand, for $E > 0$, we have the possibility of scattering. In fact, the probabilities are the same since we are only interested in either region, and even if $\psi(x)$ is non-normalizable (it wasn't even in the potential well), we can still determine the dynamics of our system for either region, assuming that either F or F' are zero.

3.6 The Finite Square Well

Let us now consider a potential well governed by

$$V(x) = \begin{cases} -V_0, & x \in [-a, a] \\ 0, & |x| > a \end{cases} \quad (49)$$

where V_0 is some positive constant. Like before, the finite square well *also* captures both scattering and bound states.

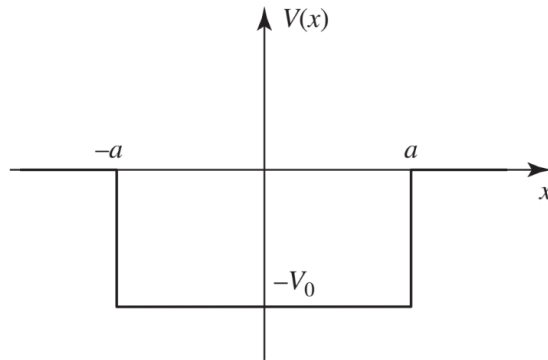


Figure 11: The finite square well.

Case $E < 0$. We will consider first the bound state. Here, we have that for $x < -a$, $V(x) = 0$, to which we can determine the SE

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = k^2\psi(x) = E\psi(x), \quad k \equiv \frac{\sqrt{-2mE}}{\hbar}$$

and, again, k is real and positive. We already know the general solution, which is $\psi(x) = A \exp(-kx) + B \exp(kx)$, but one term diverges at $x \rightarrow -\infty$, so we discard it. We thus arrive at

$$\psi(x) = B e^{kx}, \quad (x < -a).$$

Here is the difference: we must *now* consider the region $x \in [-a, a]$. In particular, we find that in this region, $V(x) = -V_0$, so the SE becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} - V_0\psi(x) = -l^2\psi(x) = E\psi(x), \quad l \equiv \frac{\sqrt{2m(E + V_0)}}{\hbar}.$$

We will briefly make a note here. It is generally *always* true that when we consider cases $E < 0$, we have that $-V_0 < E$ for some constant V_0 . In fact, we are citing the general theorem on scattering states that we found earlier.

Regardless, we find that the general solution is of the form

$$\psi(x) = C \sin(lx) + D \cos(lx), \quad x \in [-a, a].$$

For the case $x > a$, we find that the general solution is of the form $\psi(x) = F \exp(-kx) + G \exp(kx)$, which diverges for $x \rightarrow \infty$, so we also discard that term. Thus, we arrive at

$$\psi(x) = F e^{-kx}, \quad (x > a).$$

Notice, however, the solution in $x \in [-a, a]$ gives us *two* solutions: odd-parity solutions, and even-parity solutions. We will consider the even-parity solution here. We consider solutions of the form

$$\psi(x) = \begin{cases} F e^{-kx}, & x \in [a, \infty) \\ D \cos(lx), & x \in [0, a] \\ \psi(-x), & x \in (-\infty, 0] \end{cases}$$

where, notice, the bound within the potential well is *now* $x \in [0, a]$ by the even properties of our cosine term. Furthermore, since the solution is even, we can, without loss of generality, assume that the solution in the region $x < 0$ is the *same* as the solution in the region $x > 0$, with the caveat that we pick up some minus signs when exploiting our even-ness properties.

Our first boundary condition imposes continuity on the wavefunction at $x = a$, so we find that

$$F e^{-ka} = D \cos(la)$$

and the second boundary condition on the derivative imposes continuity¹⁶ such that

$$-k F e^{-ka} = -l D \sin(la).$$

We can divide each equation to rid ourselves of the constants, and we thus arrive at

$$k = l \tan(la) \tag{50}$$

¹⁶Here, we *need* continuity in the derivative since there is nothing stopping us from saying that this is not possible, unlike the delta-well, where the potential diverges at the point $x = 0$.

which is *transcendental*. This isn't a problem generally, but it is if we're trying to solve this analytically. We cannot. However, we if let

$$z \equiv la, \quad \text{and} \quad z_0 \equiv \frac{a}{\hbar} \sqrt{2mV_0}$$

we can rewrite our transcendental equation to

$$\tan(z) = \sqrt{\left(\frac{z_0}{z}\right)^2 - 1} \quad (51)$$

where $k^2 + l^2 = 2mV_0/\hbar^2$ which implies that $ka = \sqrt{z_0^2 - z^2}$.

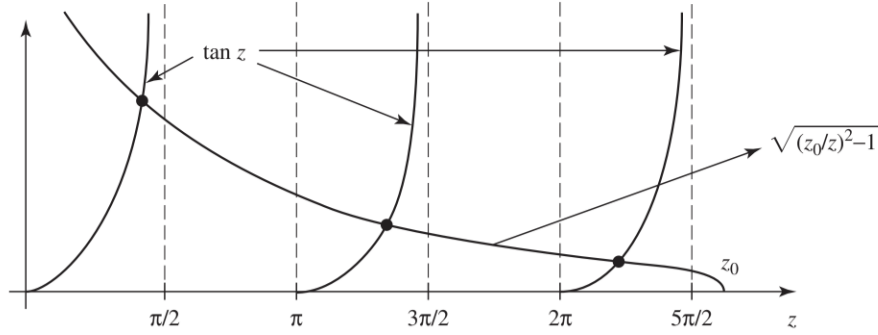


Figure 12: Graphical solution to (51) for $z_0 = 8$ (even states).

We can now look at limiting cases of our solution, which are of special interest. From (51), if z_0 is very large, that is $z_0 \gg 1$, we can see that

$$\tan(z) \approx z_0 = \frac{a}{\hbar} \sqrt{2mV_0}.$$

Here, the intersections of our transcendental equation occur just below $z_n = n\pi/2$, where n is odd. In fact, we find that

$$E_n + V_0 \approx \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}, \quad (n = 1, 3, 5 \dots). \quad (52)$$

Notice, however, that $E + V_0$ is the energy *above* the bottom of the well. In fact, we find that the right side of our equation above are the eigenstates of the infinite square well with a width of $2a$, so we expect that the finite square well becomes infinite as $V_0 \rightarrow \infty$. In this case, however, there are a *finite* number of bound states, since we also have *finite* V_0 .

Let us now consider if z_0 becomes very small. In this case, we have that fewer and fewer bound states until we reach the limiting case, where $z_0 < \pi/2$ for a single bound state. In this case, no matter how "weak" or well becomes, there will *always* be one bound state. Notice the connection to the delta-well. We have the same result as if we solved for the bound state in the potential well, so we can say that the limit case that $z_0 \ll 1$ approaches the delta-well.

Case $E > 0$. We assume the same regions and conditions as before. Here, in the region $x < -a$, we find that our general solution is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \quad (x < -a), \quad k \equiv \frac{\sqrt{2mE}}{\hbar}$$

as before. Within the well, we find

$$\psi(x) = C \sin(lx) + D \cos(lx), \quad x \in [-a, a], \quad l \equiv \frac{\sqrt{2m(E + V_0)}}{\hbar}$$

and for the right of the well, we have

$$\psi(x) = F e^{ikx} + G e^{-ikx}, \quad (x > a).$$

We must assume the same scattering state solution as before, so we will *assume* that $G = 0$ for particles coming in from the left, and so we arrive at the piecewise solution

$$\psi(x) = \begin{cases} F e^{ikx}, & x \in [a, \infty) \\ C \sin(lx) + D \cos(lx), & x \in [-a, a] \\ A e^{ikx} + B e^{-ikx}, & x \in (-\infty, -a]. \end{cases}$$

In this case, we have *four* boundary conditions. The first imposes continuity of $\psi(x)$ at $-a$, such that

$$A e^{-ika} + B e^{ika} = -C \sin(la) + D \cos(la) \quad (3.6A)$$

and the second imposes continuity of the derivative at $-a$, which yields

$$ik[A e^{-ika} - B e^{ika}] = l[C \cos(la) + D \sin(la)]. \quad (3.6B)$$

The third imposes continuity of $\psi(x)$ at a ,

$$C \sin(la) + D \cos(la) = F e^{ika} \quad (3.6C)$$

and the fourth imposes continuity of the derivative such that

$$l[C \cos(la) - D \sin(la)] = ik F e^{ika}. \quad (3.6D)$$

We must now use two of these equations to eliminate our two constants and solve for the other constants. Here, we will eliminate C and D and attempt to solve for B and F , to which we arrive at

$$B = i \frac{\sin(2la)}{2kl} (l^2 - k^2) F, \quad F = \frac{A e^{-2ika}}{\cos(2la) - i \frac{(k^2 + l^2)}{2kl} \sin(2la)}.$$

This looks horrible, but we can determine our transmission coefficient as before in Section 3.5, to which we should have

$$T^{-1} = 1 + \frac{V_0^2}{4E(E + V_0)} \sin^2 \left[\frac{2a}{\hbar} \sqrt{2m(E + V_0)} \right] \quad (53)$$

and the reflection coefficient is $R = 1 - T$, such that

$$R = - \frac{V_0^2 \sin^2[2a \sqrt{2m(E + V_0)}/\hbar]}{4E(E + V_0) + V_0^2 \sin^2[2a \sqrt{2m(E + V_0)}/\hbar]}. \quad (54)$$

/ Notice, that whenever the sine is zero, we have our energy eigenstates for the infinite square well, such that

$$\frac{2a}{\hbar} \sqrt{2m(E + V_0)} = n\pi \implies E_n + V_0 = \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}.$$

In this case, our well effectively becomes "transparent", such that the particle will *always* escape out of the well. In fact, we find that $T = 1$ and so $R = 0$, as we expect.

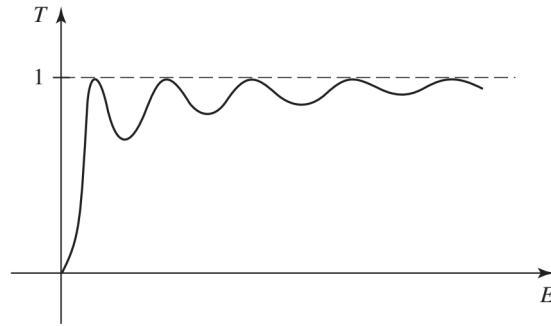


Figure 13: Transmission coefficient as a function of energy.

4 Preview of Spin

In classical mechanics, we have two kinds of angular momentum. **Orbital angular momentum** refers to the angular momentum referenced about some object; **spin angular momentum**, on the other hand, refers to the angular momentum due to rotation of the object about some axis. In the classical picture, these are the same. Spin arises from the orbital angular momentum of all the infinitesimal masses of the rigid body.

Quantum mechanically, these are *not* the same. Spin is an *intrinsic* property of quantum particles, and although it has units of angular momentum, it is **not** actually rotating about an internal axis. Furthermore, we know that spin is a quantized quantity. The difference between the quantum picture and the classical picture is *extremely* important. We will attempt to rectify it here.

4.1 The Stern-Gerlach Experiment

Let us begin by supposing we are attempting to conduct an experiment. We assume to have an experimental apparatus as in Fig. 14.

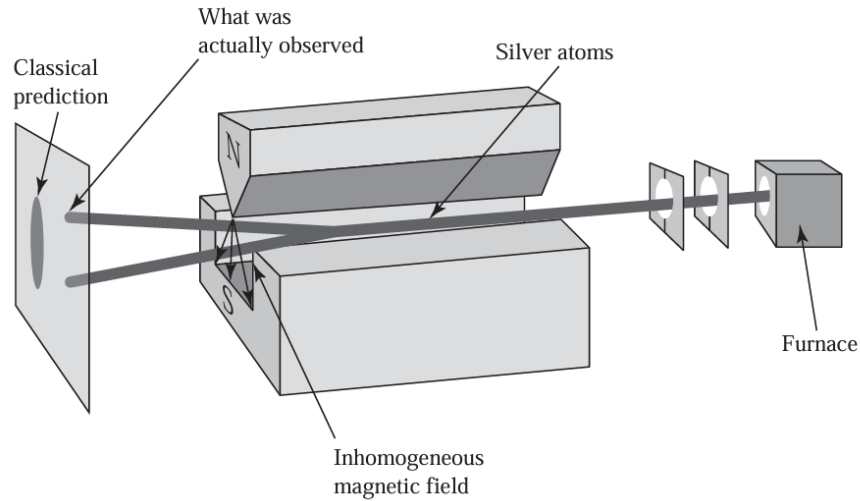


Figure 14: The Stern-Gerlach Apparatus

In particular, we are interested in what happens if we heat silver (Ag) atoms and shoot them through a collimator under the influence of an inhomogeneous magnetic field. That is, the magnetic field varies as a function of space.

Classically, we can make our prediction: the silver atom is a neutral atom (it has an equal number of protons and electrons), possessing a magnetic moment $\vec{\mu}$. In this way, we can define other quantities of the atom, such as its torque due to the magnetic field,

$$\mathbf{N} = \vec{\mu} \times \mathbf{B},$$

its energy subject to the magnetic field,

$$E_{\mathbf{B}} = -\vec{\mu} \cdot \mathbf{B}$$

and the corresponding magnetic force,

$$\mathbf{F}_B = -\vec{\nabla} E_B = \frac{\partial}{\partial z} [\vec{\mu} \cdot \mathbf{B}] \approx \mu_z \frac{\partial B_z}{\partial z}.$$

where we approximate the dot product since we assume much of the magnetic moment is oriented about the z -direction. Naturally, we can make some predictions about the proportionalities of the spin of the atom. In fact, we can argue that

$$\mathbf{S} \propto \vec{\mu}, \quad S_z \propto \mu_z, \quad \mathbf{F}_z \propto S_z \frac{\partial B_z}{\partial z}.$$

What does this say about our expected result? If we consider our atom, and therefore electrons, to be like a classically spinning object, then we can consider every possible orientation of our electron the *moment* it enters the apparatus. We would then find a range of possible outcomes, and thus a *distribution* of atoms appearing on our observation screen.

However, what *actually* happens is that we find our "distribution" to be quantized. That is, we see only *two* deflections at the points $\hbar/2$ and $-\hbar/2$ due to the spin angular momentum instead. This result, in turns out, is a general consequence of the type of particles we are dealing with, called **spin 1/2 particles**.

A The Dirac Delta Function

Many of the useful relations utilize the Dirac delta function. However, the Dirac delta is not a *true* function. It is an example of a generalized function, which specifically acts on the real numbers. In particular, the Dirac delta is represented heuristically as

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \quad (\text{A.1})$$

such that it has a value of zero everywhere in $\mathbb{R}$, except for $x = 0$ itself.

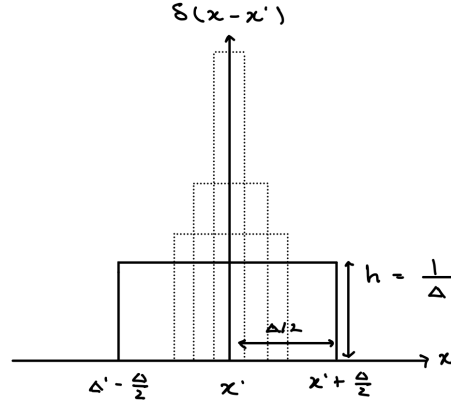


Figure A.1: Partitioning an approximation of the Dirac delta into limiting rectangles.

The Dirac delta is also constrained to satisfy the identity

$$\int_{-\infty}^{\infty} \delta(x) dx = 1.$$

Under a Fourier transform, one can also show that the Dirac delta, centered at some $x_0 = \alpha$, is given by

$$\delta(x - \alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ip(x-\alpha)} dp.$$

With this understanding of the Dirac delta, we can extend these ideas to arbitrary functions. The most commonly used is its sifting property, which says that the integral of any function multiplied by the time-delayed Dirac delta is

$$\int_{-\infty}^{\infty} f(\tau) \delta(\tau - t) d\tau = f(t). \quad (\text{A.2})$$

Similarly, the convolution of a function, $f(T)$ on $\delta(T - t)$ translates $f(T)$ by the same amount, t :

$$[f(T) * \delta(T - t)] = \int_{-\infty}^{\infty} f(T) \delta(T - t - \tau) d\tau = \int_{-\infty}^{\infty} f(T) \delta(\tau - T + t) d\tau = f(T - t) \quad (\text{A.3})$$

where we have used the relation $\delta(t) = \delta(-t)$. We now state three more (relatively) important properties of the Dirac delta.

Scaling. The Dirac delta satisfies the following scaling property for some non-zero scalar, α .

$$\delta(\alpha x) = \frac{\delta(x)}{|\alpha|} \quad (\text{A.4})$$

Proof. We have

$$\int_{-\infty}^{\infty} g(x) \delta(\alpha x) dx = \frac{1}{\alpha} \int_{-\infty}^{\infty} g\left(\frac{x'}{\alpha}\right) \delta(x') dx' = \frac{g(0)}{\alpha}$$

for the change in variables $x' = \alpha x$. By similar methods, if $\alpha < 0$, that is $-\alpha = |\alpha|$, then

$$\int_{-\infty}^{\infty} g(x) \delta(-\alpha x) dx = \frac{g(0)}{-\alpha}$$

and so we arrive at (A.4). □

Symmetry. The Dirac delta is an even distribution, in the sense that

$$\delta(-x) = \delta(x) \quad (\text{A.5})$$

as used for (A.3). This can be seen in Fig. (A.1).

Distributional Product. In the most general sense, the product of $\delta(x - \alpha)$ for some $(x - \alpha)^n$ is zero.¹⁷

$$(x - \alpha)^n \delta(x - \alpha) = 0. \quad (\text{A.6})$$

Heaviside Step Derivative. The final most important property involves the derivative of the Dirac delta. In particular, the relation is given by

$$\frac{dH(x - \alpha)}{dx} = \delta(x - \alpha) \quad (\text{A.7})$$

for the Heaviside function centered at α , $H(x - \alpha)$.

Proof. Let us recall (A.7). Then,

$$\int_{-\infty}^{\infty} \frac{dH(x - \alpha)}{dx} dx = \lim_{s \rightarrow -\infty} \lim_{v \rightarrow \infty} \left[H(x - \alpha) \right]_s^v = H(v - \alpha) - H(s - \alpha) = 1$$

since the Heaviside is zero for $s \rightarrow -\infty$. However, notice that there exists a discontinuity at $x = \alpha$, namely that $H(x = \alpha) \rightarrow \infty$, due to the jump discontinuity. We can see that

$$H(x - \alpha) = \begin{cases} 0, & x \neq \alpha \\ \infty, & x = \alpha \end{cases}$$

and comparing this result whilst taking note of the fact that the integral of $H(x - \alpha)$ over all space is 1, we arrive at

$$\delta(x - \alpha) = \frac{dH(x - \alpha)}{dx}.$$

□

¹⁷I'm omitting the proof for this since it seems complicated.

B Generalized Ehrenfest's Theorem

You may recall Ehrenfest's theorem in Chapter 1.6.2. Ehrenfest's theorem, in fact, is a consequence of a more general statement about the time derivative of the expectation value of an operator.

Proof. To see this, let us suppose that a system is determined by a wavefunction, $\psi(\mathbf{r}, t)$, with an associated operator, $\hat{A}$. The time derivative of the operator is such that

$$\begin{aligned} \frac{d\langle \hat{A} \rangle}{dt} &= \frac{d}{dt} \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \hat{A} \psi(\mathbf{r}, t) d\mathbf{r} \\ &= \int_{-\infty}^{\infty} \left[\frac{\partial \psi^*(\mathbf{r}, t)}{\partial t} \right] \hat{A} \psi(\mathbf{r}, t) d\mathbf{r} + \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \left[\frac{\partial \hat{A}}{\partial t} \right] \psi(\mathbf{r}, t) d\mathbf{r} + \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \hat{A} \left[\frac{\partial \psi(\mathbf{r}, t)}{\partial t} \right] d\mathbf{r}. \end{aligned}$$

But $\int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \left[\frac{\partial \hat{A}}{\partial t} \right] \psi(\mathbf{r}, t) d\mathbf{r}$ is simply the expectation value of the $\hat{A}$ with respect to time. Then,

$$\frac{d\langle \hat{A} \rangle}{dt} = \int_{-\infty}^{\infty} \left[\frac{\partial \psi^*(\mathbf{r}, t)}{\partial t} \right] \hat{A} \psi(\mathbf{r}, t) d\mathbf{r} + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle + \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \hat{A} \left[\frac{\partial \psi(\mathbf{r}, t)}{\partial t} \right] d\mathbf{r}.$$

However, Schrödinger's equation says that

$$\frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \frac{1}{i\hbar} \mathcal{H} \psi(\mathbf{r}, t), \quad \frac{\partial \psi^*(\mathbf{r}, t)}{\partial t} = -\frac{1}{i\hbar} \psi^*(\mathbf{r}, t) \mathcal{H}$$

and substituting these values back into the time derivative of $\hat{A}$,

$$\begin{aligned} \frac{d\langle \hat{A} \rangle}{dt} &= \frac{1}{i\hbar} \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \hat{A} \mathcal{H} \psi(\mathbf{r}, t) d\mathbf{r} - \frac{1}{i\hbar} \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) \mathcal{H} \hat{A} \psi(\mathbf{r}, t) d\mathbf{r} + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle \\ &= \frac{1}{i\hbar} \int_{-\infty}^{\infty} \psi^*(\mathbf{r}, t) [\hat{A} \mathcal{H} - \mathcal{H} \hat{A}] \psi(\mathbf{r}, t) d\mathbf{r} + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle. \end{aligned}$$

Since $\hat{A} \mathcal{H} - \mathcal{H} \hat{A}$ is the commutation relation between $\hat{A}$ and $\mathcal{H}$, the most general form is then,

$$\boxed{\frac{d\langle \hat{A} \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{A}, \mathcal{H}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle.} \quad (\text{B.1})$$

The proof, however, assumes that the Hamiltonian operator, $\mathcal{H}$, is Hermitian. $\square$

Example B.1. Suppose a wavefunction, $\psi = \psi(x, t)$, in 1D is confined to a one-dimensional system with a potential $V = V(x, t)$. Derive the instantaneous expectation value of the momentum with respect to time.

First, note that the Hamiltonian in a 1D potential is given by

$$\mathcal{H} = \frac{\hat{P}^2}{2m} + V(x, t).$$

Since the operator of interest is $\hat{A} = \hat{P} = -i\hbar\partial^2/\partial x^2$, the expectation value of the commutator relation is

$$\langle \hat{P}, \mathcal{H} \rangle = \left\langle \left[\hat{P}, \frac{\hat{P}^2}{2m} + V(x, t) \right] \right\rangle = \frac{1}{2m} \langle [\hat{P}, \hat{P}^2] \rangle + \langle [\hat{P}, V(x, t)] \rangle = \langle [\hat{P}, V(x, t)] \rangle$$

since the operator, $\hat{P}$ commutes with itself, such that $[\hat{P}, \hat{P}^2] = 0$. Furthermore, $\hat{P}$ has no time dependence, and so $\partial\hat{P}/\partial t = 0$. Then,

$$\begin{aligned} \frac{d\langle \hat{P} \rangle}{dt} &= \frac{1}{i\hbar} \langle [\hat{P}, V(x, t)] \rangle \\ &= \int_{-\infty}^{\infty} \psi^*(x, t) V(x, t) \frac{\partial}{\partial x} \psi(x, t) dx - \int_{-\infty}^{\infty} \psi^*(x, t) \frac{\partial}{\partial x} [V(x, t) \psi(x, t)] dx \end{aligned}$$

where we have replaced $\hat{P}$ with $-i\hbar d/dx$. Product rule on the second term shows that

$$\frac{\partial}{\partial x} [V(x, t) \psi(x, t)] = V(x, t) \frac{\partial \psi(x, t)}{\partial x} + \psi(x, t) \frac{\partial V(x, t)}{\partial x}.$$

We can see that the second term cancels with the first term in the expectation value of $d\langle \hat{P} \rangle/dt$, and so the equation simplifies to

$$\frac{d\langle \hat{P} \rangle}{dt} = - \int_{-\infty}^{\infty} \psi^*(x, t) \left[\frac{\partial V(x, t)}{\partial x} \right] \psi(x, t) dx = \left\langle - \frac{\partial V(x, t)}{\partial x} \right\rangle = \langle F \rangle.$$

Example B.2. Use the result of Example B.1 to determine $\langle \hat{P} \rangle$ as a function of t in the case of a quantum harmonic oscillator. You may leave unevaluated integrals in your solution.

The potential for the quantum harmonic oscillator in 1D is $V(x) = m\omega^2 \hat{X}^2/2$. Taking the derivative with respect to x ,

$$\frac{dV(x)}{dx} = \frac{d}{dx} \left[\frac{1}{2} m\omega^2 \hat{X}^2 \right] = m\omega^2 \hat{X}.$$

Rearranging the result of (B.1) to be in terms of $\hat{P}$ and taking the integral for the time $t' \in [0, t]$,

$$\int_0^t \left[\frac{d\langle \hat{P} \rangle}{dt} \right] dt = - \int_0^t \left\langle \frac{dV(x)}{dx} \right\rangle dx \implies \langle \hat{P} \rangle = \langle \hat{P}(0) \rangle - m\omega^2 \int_0^t \langle \hat{X}(t') \rangle dt'.$$

Then, since $\hat{X}(t') = x(t')$, we arrive at

$$\langle \hat{P}(t) \rangle = \langle \hat{P}(0) \rangle - m\omega^2 \int_0^t \langle x(t') \rangle dt'.$$

In particular, notice that the expectation value of momentum in this form is exactly analogous to integrating Hamilton's equation of motion in $\dot{p}(t)$ for a harmonic oscillator.

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